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RESEARCH ARTICLE

The ERA5 global reanalysis from 1940 to 2022

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Abstract

We provide a description and concise evaluation of the European Centre of Medium-range Weather Forecasts Reanalysis v.5 (ERA5) global reanalysis from an additional extension back to 1940 that was released in March 2023, including its timely updates to the end of 2022. The ERA5 product from 1979 to end 2020 and a preliminary back extension from 1950 to 1978 have already been described elsewhere. The new back extension that spans 1940 to 1978 represents the official release and supersedes the preliminary product. Currently, the ERA5 data record extends over more than 83 years of hourly global three-dimensional fields for many quantities that describe the global atmosphere, land surface, and ocean waves at a horizontal resolution of about 31 km. ERA5 relies on the ingestion of sub-daily in-situ and satellite observations, and the number of these increases from 17,000 per day in 1940 to 25 million per day by 2022. Accordingly, the quality of the reanalysis improves throughout the period. Over the Northern Hemisphere ERA5 generally provides a reliable representation of the synoptic situation from the early 1940s and provides long-term variability that is in line with other datasets. Over the Southern Hemisphere, however, for the early period the description of ERA5 seems mainly statistical. Furthermore, there is a small deviation in surface temperature compared with reconstructions based on monthly aggregations of observations over land before 1946. For this period, the absence of upper air temperature observations reveals a model cold bias in the lower stratosphere. For the period from 1950 to 1978, the final release described here improves on the suboptimal treatment of International Best Track Archive for Climate Stewardship observations in the preliminary release, with, as a result, a much more homogeneous representation of tropical cyclones over the entire ERA5 record. Longer spin-up periods also have a beneficial impact on soil moisture.

KEYWORDS

climate reanalysis, data assimilation, ECMWF, ERA5

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1 | INTRODUCTION

This article provides an overview of the Copernicus Climate Change Service (C3S; Buontempo *et al.*, 2022; Thépaut *et al.*, 2018) European Centre of Medium-range Weather Forecasts (ECMWF) fifth-generation global reanalysis, ERA5 (Hersbach *et al.*, 2020), continued in near real-time, and now extended to cover the time period since the year 1940.

After Hersbach *et al.* (2020) documented the ERA5 configuration and the performance of ERA5 for the years from 1979 onwards, Bell *et al.* (2021) presented the initial evaluation of a preliminary extension back to 1 January, 1950, hereafter referred to as ERA5.P. Based on these results, several improvements were made to produce the final segment covering the years 1950–1978, as well as extend the ERA5 reanalysis back to the year 1940. This article explains these improvements and provides an overall evaluation of the ERA5 performance for the time period from 1 January 1940, to 31 December 2022.

One particular interest of this reanalysis is to cover a new phase of the Anthropocene (Witze, 2023) started in the early 1950s, a time of atmospheric nuclear weapons testings and acceleration of industrialization and fossil fuel consumption. This article presents possible indications that a marker of this time period is also present in atmospheric records.

Consistent with the early articles that proposed concepts of reanalyses (Bengtsson & Shukla, 1988; Trenberth & Olson, 1988)—that is, for two different reasons—reanalyses generally strive to serve two seemingly irreconcilable applications. A first objective is to achieve maximum synoptic realism for any given date, by assimilating all available observations. A second goal is to generate gridded products for climate research; that is, with consistency and continuity over time. However, the more the observation quality and quantity change over time, the more difficult it is to achieve this second goal. Indeed, changes in the observing system are facts that cannot be undone, at least by any method known at present. The reason is that there is no single component of the observing system that has remained untouched over the years, whether that be in the instrumentation methodology, data transmission method, or actual station coverage. The same remark applies to the gridded information used for forcing or boundary conditions in reanalyses. Even advanced attempts to circumvent some of this problem, such as centennial reanalyses assimilating only surface observations (e.g., Compo *et al.*, 2011), still make use of satellite information at some point (i.e., collected in the recent times) in order to derive horizontal structure functions for the sea-surface temperature (SST), which are then used for the pre-satellite era (e.g., Rayner *et al.*, 2003). In addition, there are also indications that observation measurement quality (for surface pressure) improved over time (e.g., Poli *et al.*, 2016).

In ERA5, no attempt was made to restrict the observation input in such a way. Instead, the approach comprises confronting the problem by weighting the information according to its quality, with an aim of informing users so they can make appropriate decisions in their applications. The changes in the observing system are numerous. The conventional part of the observing system is supported by aircraft, balloon-borne instruments, and surface-based stations or platforms such as buoys and ships. From these reports received, only the following observables are assimilated: temperature, humidity, pressure, wind, and snow depth. Figure 1 shows the spatio-temporal evolution of this coverage, by observable and by vicennium (20-year period). In the first vicennium, the observations assimilated primarily comprise surface pressure and wind reports, most notably located over the Northern Hemisphere continents. In comparison, the southern oceans are only marginally covered, with many areas featuring on average less than one report per month. Upper air observations are also nearly exclusively located over Northern Hemisphere continents, with anecdotal coverage over oceans.

The situation improves notably in the next vicennium with the generalization of radiosonde observations, following the International Geophysical Year (1957-1958), supplemented afterwards by an increasing number of aircraft reports. In the most recent years, reports of surface pressure (or equivalent geopotential height over mountains) are most numerous in regions where they can be provided by automatic stations or buoys. Data gaps remain, however, over Antarctica, Greenland, central Africa, northern Siberia, and the Himalayas.

Such monumental changes in the observation coverage cannot be without consequence on the reanalysis products: varying amounts of observation modulate the biases that are inevitably present in the various ingredients of a reanalysis (observations, models and their forcings) as discussed by, for example, Simmons *et al.* (2021) for a few key variables.

For the remotely sensed part of the observing system, the coverage in ERA5 starts in 1970 with ozone, followed by infrared radiances in 1972 as described by Bell *et al.* (2021). Figure 2 shows the spatial distribution, by assimilated observable. The figure includes radar rain rates over the United States, but otherwise all observations on this figure come from satellites. With the exception of ozone and Global Navigation Satellite System radio occultation, usage of satellite data is greater over oceans than over land, for reasons that differ by observing technique.

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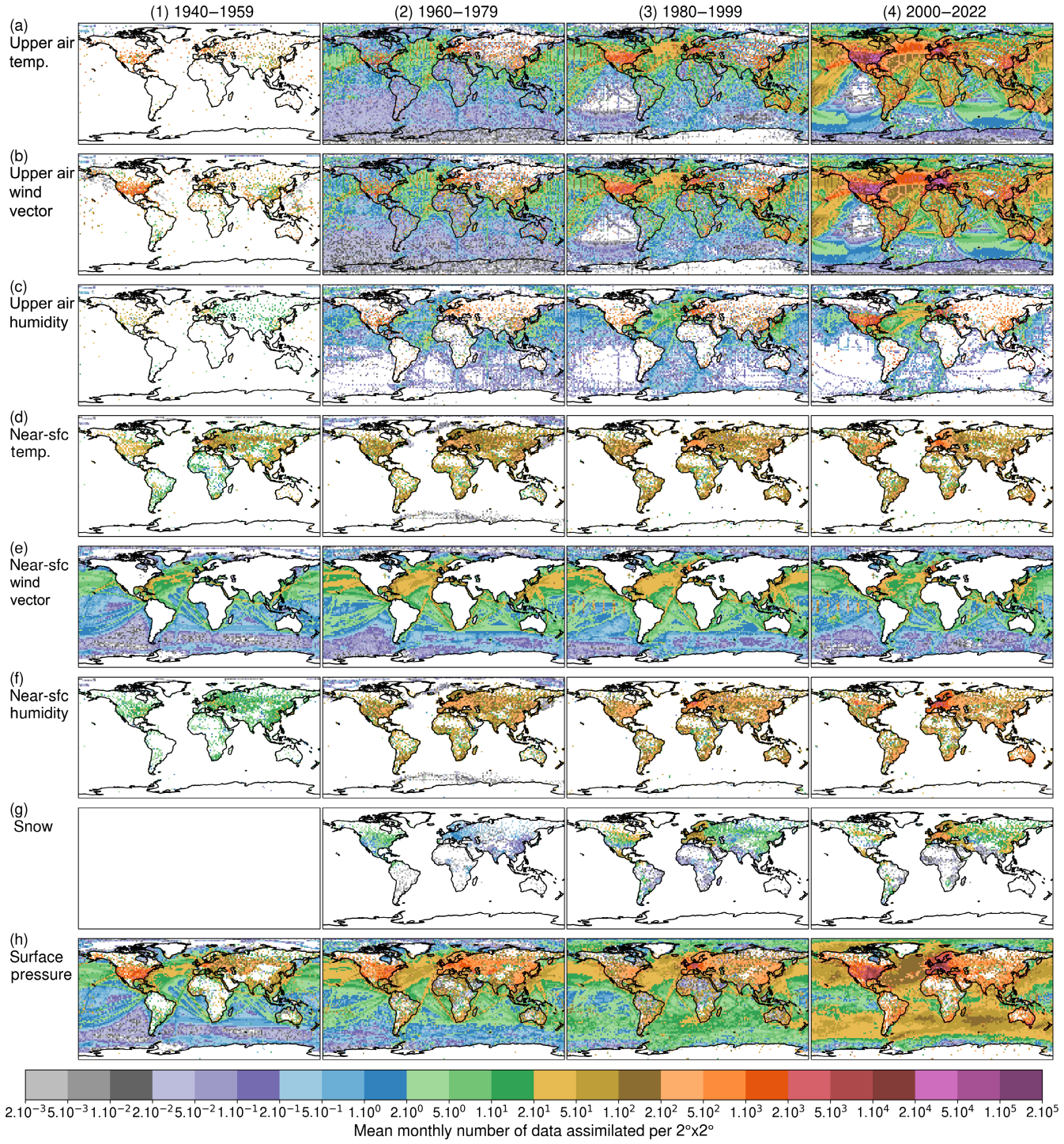

FIGURE 1 Maps of data counts at 2° × 2° (latitude, longitude) resolution, for the conventional part of the observing system, separating by vicennium (20-year time period, from left to right, except for the last column, which shows observations until December 2022) and by observable (from top to bottom). Numbers indicate the average data counts assimilated per month. White areas indicate no data assimilated.

European contributions to the global observing system, beyond European land and water areas, are found in both figures. One may note in particular satellite observations of Africa, western and central parts of Asia, and the Indian Ocean from geostationary orbit (EUMETSAT), as well as in-situ observations outside Europe by aircraft and ship radiosoundings for the upper air and by ships and buoys for the marine surface (EUMETNET).

Information about the evolution of the observing system is of direct relevance to all aspects of performance

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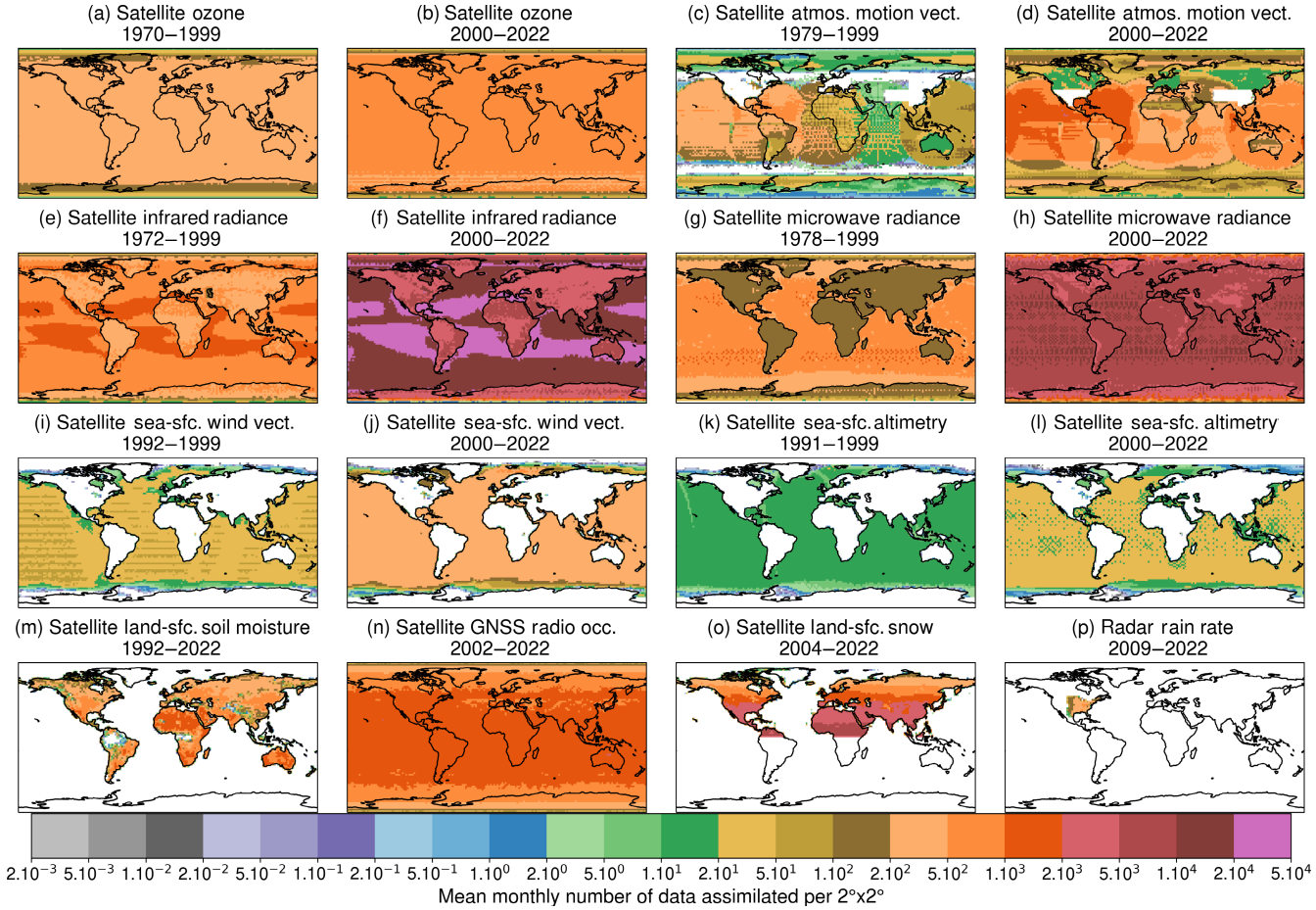

FIGURE 2 Maps of data counts at 2° × 2° (latitude, longitude) resolution, for the remotely sensed observations, separating by observable and further splitting into two periods for some of the longest-running observing system components. Numbers indicate the average data counts assimilated per month. White areas indicate no data assimilated.

discussed in this article. This information should also be of interest to users whose applications may present a sensitivity to changes in the observational constraints.

The resulting ERA5-analysed model fields are available to users through the C3S Climate Data Store for the period 1 January, 1940, to present. This represents more that 83 years of hourly estimates of the global atmosphere, land surface, and ocean waves, at a horizontal resolution of 31 km for the high-resolution (HRES) product, a significant advance with respect to the predecessor ECMWF global reanalysis ERA-Interim, at 78 km horizontal resolution (Dee *et al.*, 2011). In addition, ERA5 uncertainty estimates are provided by a 10-member ensemble of data assimilations (EDA), at half the horizontal resolution and three-hourly temporal resolution. For almost all quantities, precalculated monthly mean values are available as well.

The ERA5.P evaluation had identified suboptimal aspects in the preliminary extension to 1950. One aspect was the inadequate assimilation of six-hourly tropical cyclone best-track pressure reports from the International Best Track Archive for Climate Stewardship (IBTrACS; Knapp *et al.*, 2010). The most notable consequence for ERA5.P was the presence of unrealistically deep-pressure lows, sometimes significantly offset from the position of the pressure minimum indicated in IBTrACS, associated in turn with overestimated wind speeds and significant wave heights. Another aspect was a limited pool of surface pressure observations subject to bias corrections. The final production resolves these two issues. The methodology and results are presented in Section 2.

The subsequent sections discuss various key aspects of the performance of the 1940–2022 ERA5 data record. Section 3 presents the performance at synoptic time-scales. Sections 4 and 5 present the performance of ERA5 in representing low-frequency variability in (respectively) the water cycle and the thermal structure of the atmosphere. The representation of temperature extremes is also briefly discussed. For these two latter sections the results shown in this article for 2000–2006 are based on ERA5.1 (Simmons *et al.*, 2020). ERA5.1 has generally smaller temperature biases than ERA5 in the lower stratosphere and upper

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troposphere for this period. It also provides some improvements to the analysis of ozone and the quasi-biennial oscillation of tropical stratospheric winds. ERA5.1 is very close to ERA5 in the lower and middle troposphere, and observation usage is very similar, in general.

Finally, Section 6 presents conclusions and perspectives for future work.

2 | PRODUCTION OF ERA5 FROM 1940 TO 1978

This section describes the production of the ERA5 segment from 1940 to 1978. Besides an additional decade from 1940 to 1949, this includes an improvement of the preliminary ERA5.P product for 1950–1978. Although some key aspects of innovation for that period are given here, a more systematic evaluation of ERA5 and its comparison with ERA5.P is presented in Sections 4 and 5.

2.1 | Production aspects

The back extension from 1940 to 1978 was produced in two separate phases of around 20 years, with each running four streams of about 5 years in parallel. As for the other parts of ERA5, streams contained an HRES deterministic component (at TL639, or about 31 km horizontal resolution) and an EDA (at TL319, or about 63 km). The latter not only provides the ERA5 uncertainty estimate but also the flow-dependent background error covariance matrix that is used in the ERA5 atmospheric data assimilation component (four-dimensional variational, 4D-Var). More information on the ERA5 configuration and assimilation system can be found in Hersbach *et al.* (2020, section 2).

These production streams were later consolidated into one product that was added to the existing catalogue entries in the C3S Climate Data Store. Details of these eight (two times four) production streams are provided in Table 1 and Figure 3. Transition points between consolidated streams are also indicated by the red (ERA5) and blue (ERA5.P) vertical lines in Figure 4. All ERA5 streams benefited from the improved surface pressure bias scheme and revised assimilation of IBTrACS observations, as described later in this section.

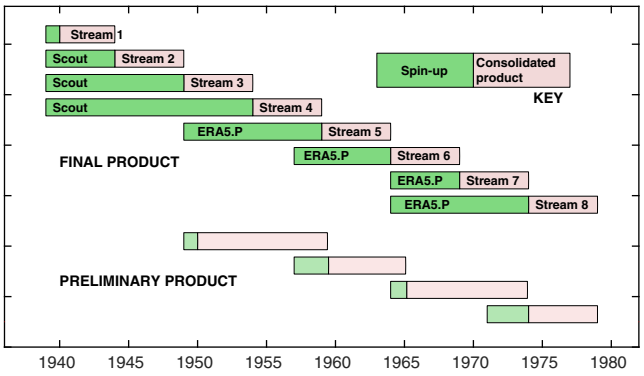

FIGURE 3 European Centre of Medium-range Weather Forecasts Reanalysis v.5 (ERA5) production streams listed in Table 1, showing spin-up and consolidated production periods. Also shown (fainter colours, bottom bars) are the production streams for the preliminary back extension (ERA5.P), illustrating the longer temporal coverage of the final product, as well as the longer spin-up periods for all final product streams.

TABLE 1 Details of the eight production streams, comprising both high-resolution (HRES) and ensemble of data assimilations (EDA), from 1940 to 1978 that were merged into the consolidated public product—details of the European Centre of Medium-range Weather Forecasts Reanalysis v.5 (ERA5) system can be found in Hersbach *et al.* (2020, section 2).

Stream	Used for years	Start date	Effective initial state	Effective spin-up
1	1940–1943	2 January 1939	1 January 1939, Scout	1–5 years
2	1944–1948	15 October 1943	1 January 1939, Scout	5–10 years
3	1949–1953	15 October 1948	1 January 1939, Scout	10–15 years
4	1954–1958	15 October 1953	1 January 1939, Scout	15–20 years
5	1959–1963	14 December 1958	1 January 1949, ERA5.P	10–15 years
6	1964–1968	14 December 1963	1 January 1957, ERA5.P	7–12 years
7	1969–1973	14 December 1968	1 January 1964, ERA5.P	5–10 years
8	1974–1978	14 December 1973	1 January 1964, ERA5.P	10–15 years

Note: Start dates are for the EDA components (start dates for the HRES are 1 day later). Indicated periods run from 1 January to 31 December inclusive. Streams 5–8 were produced and made available prior to streams 1–4 and were initiated from production streams of ERA5.P (ERA5 preliminary extension back to 1 January 1950)—as listed in Bell *et al.* (2021, tab. 1). Streams 1–4 were initialized from a deterministic "scout" reanalysis at EDA resolution (TL319) that was produced ahead as one continuous stream from 1939 onwards. The spin-up of these initialized conditions considerably extends the effective spin-up period of the ERA5 consolidated product, such that smoother transitions between streams are obtained for soil moisture and stratospheric humidity.

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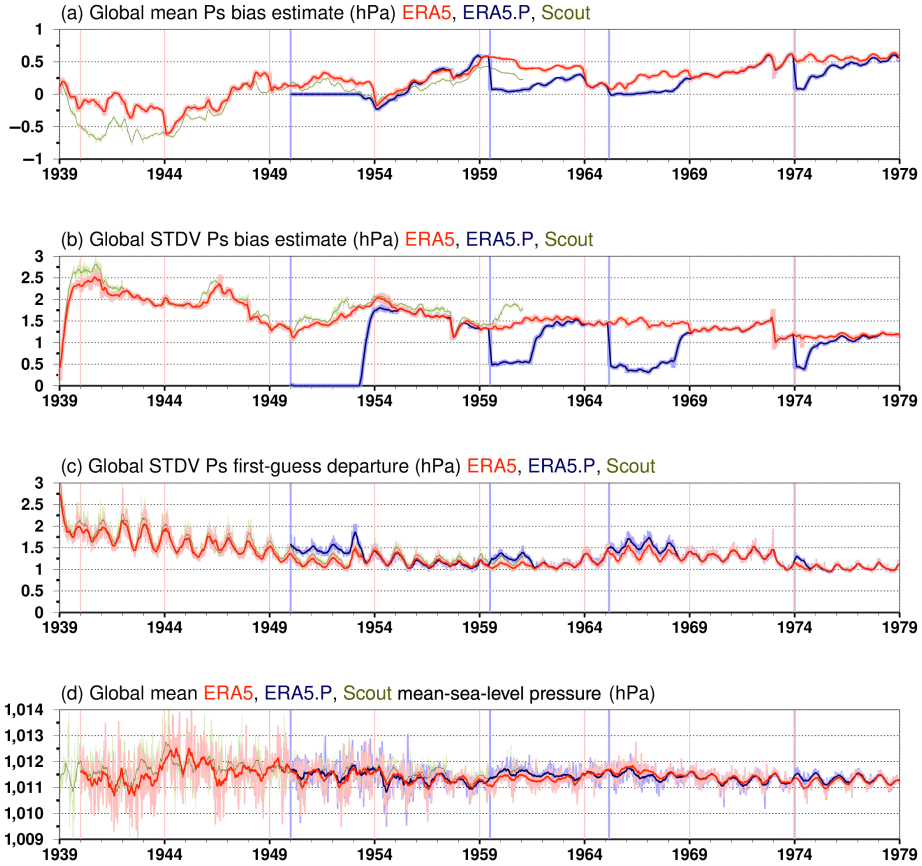

FIGURE 4 Daily (light colours) and 30-day running mean (dark colours) of the (a) mean and (b) standard deviation (STDV) of the estimated bias, and (c) STDV of the departure of the observation from the model first guess for surface and mean-sea-level pressure observations averaged over all observation locations and (d) mean of the model mean-sea-level pressure integrated over the entire globe and averaged over 0000, 0600, 1200, and 1800 UTC for ERA5 (red), ERA5.P (blue), and the scout (green) that was used to initialize ERA5 for the first four streams. Transition points of the ERA5 and ERA5.P production streams are indicated by vertical lines in corresponding colours. ERA5, European Centre of Medium-range Weather Forecasts (ECMWF) Reanalysis v.5; ERA5.P, ERA5 preliminary extension back to 1 January 1950.

The phase that covered 1959 to 1978 inclusive was produced first. Each stream (numbered from 5 to 8 in this article) included a short spin-up period from an initial state from original ERA5.P analyses (15 December and 14 December of the previous year for the HRES and EDA components respectively). The timing of these initial states was chosen to be as late as possible in ERA5.P production streams such that ERA5 could benefit from a maximum amount of effective spin-up time that had been accumulated already by ERA5.P.

The phase that covered 1940 to 1958 inclusive (streams 1–4) was produced after the production phase for 1959 had completed. For the streams that started in the 1940s, no initial conditions were available from original ERA5.P analyses, nor from ERA-Interim or ERA-40. The same strategy could have been followed here as for the initial stream of ERA5.P; that is, initializing from ERA-PreSAT (Hersbach *et al.*, 2017). However, this had been found sub optimal leading to larger spin up time, especially for humidity and soil moisture.

As an alternative it was chosen to initialize streams 1–4 from a lower resolution test run that had been conducted earlier. That "scout" involved one long continuous stream from 1939 to 1958. It contained one single deterministic reanalysis (i.e., no EDA) at the reduced resolution of the EDA (i.e., TL319 rather than TL639). It used a purely climatological background error covariance matrix rather than from an EDA, since the latter was unavailable. Its much lower cost made it affordable to run this scout run in parallel with the first production phase (1959–1978) and also to let it progress faster such that it was available when required. The scout also included the aforementioned and subsequently to be described improvement for surface pressure bias correction and assimilation of IBTrACS. For the scout the initial state of 1 January 1939, was taken from an initial ERA5.P state valid for 1 January 1959. Since this initial state originates from an ERA5.P stream that was started for January 1949—see Bell *et al.* (2021, tab. 1)—it has a 10-year spin-up period for the long-term mean state, which is particularly beneficial for soil moisture. For the troposphere at least, this erroneous, initial synoptic state was corrected by the data assimilation system within a few weeks and had converged to the best possible estimate well before 1940 when it mattered.

Each of the 1940–1958 streams was initialized from the scout and therefore benefited from its long spin-up (see Table 1 and Figure 3). Stream 1 was started from 2 January (EDA) and 3 January (HRES) 1939, and streams 2, 3, and 4 were started from 15 October (EDA) and 16 October (HRES) 1943, 1948, and 1953 respectively.

As will be shown later, the much improved spin-up periods for this new ERA5 extension allowed for a more continuous end product than ERA5.P had provided.

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2.2 | Surface pressure bias correction

As described in Bell *et al.* (2021), ERA5.P suffered from an error in the correction scheme that is in place to handle biases in surface and mean-sea-level pressure observations—VarBC; see Hersbach *et al.* (2020, section 2.2). ERA5 uses such observations from sources that had been used before by ERA-40 (Uppala *et al.*, 2005) and ERA-Interim (Dee *et al.*, 2011), called ERA-40 BUFR data in Bell *et al.* (2021), and from the International Surface Pressure Databank (ISPD, version 3.2.6) and the International Comprehensive Ocean–Atmosphere Data Set (ICOADS, version 2.5.1) that were not used. Around halfway through the production of ERA5.P it was discovered that the pressure bias correction scheme had only been applied to ERA-40 BUFR observations. Since these were not available prior to September 1957, it meant that for that period, up until that point in production, no bias corrections had been applied to pressure observations at all. It was decided to correct this during the production, which meant that the bias scheme was not applied uniformly to all pressure observations in ERA5.P.

This was rectified in the ERA5 production described here; that is, VarBC was also used throughout for ISPD and ICOADS. The improvement of statistics is shown in Figure 4, which shows the time evolution of the mean (Figure 4a) and standard deviation (STDV; Figure 4b) of the estimated bias and the STDV of the departure from the model first guess (Figure 4c) over all pressure observations that were assimilated in a particular 12-h assimilation window. Results are in red for ERA5 and in blue for ERA5.P. VarBC needs several months of spin-up, and for this reason the initial bias estimates of the eight production streams (thin vertical red lines) used the corresponding values from the initializing states; that is, from the scout (light green curve) for streams 1–4 and from ERA5.P for streams 5–8. The issue in ERA5.P can be seen clearly; prior to 29 April 1953—Bell *et al.* (2021) inadvertently stated 1955—no bias correction is applied, and for the first part of later streams estimates are only applied to part of the assimilated observations. Large jumps appear at stream transitions (indicated by vertical blue lines). For ERA5 the evolution is much more consistent. The initialization from both the scout and ERA5.P for streams 1–4 and 5–8, works well in general. ERA5 overlaps well with periods where ERA5.P had the correct scheme in place for a sufficiently long period. Note, however, that there are some deviations in the evolution of the scout mean bias correction and ERA5, which does lead to some jumps for the latter for 1 January 1944, 1949, and 1954. There are no jumps for STDV. Figure 4c clearly shows the adverse effect on first-guess departures for ERA5.P in the epochs when it did not account for bias corrections for all pressure observations and the evolution for ERA5 is more continuous. Departures for the scout are slightly worse.

Figure 4d shows the evolution of the mean, model mean-sea-level pressure, integrated over the entire globe (i.e., not just at observation locations). From this it emerges that ERA5 and ERA5.P compare rather well, with the largest differences generally occurring for epochs when bias estimates (shown in Figure 4a,b) differ most as well. The difference between ERA5 and the scout seems larger. This applies especially to the 1940s, which is also the period when the global mean ERA5 mean-sea-level pressure is most volatile, probably due to a larger variation of this quantity due to the sparser controlling observation network. Apparently, the differences in resolution (31 km versus 63 km) and background error covariance (EDA based versus static), between ERA5 and the scout, have a larger impact on the variability of the model global mean pressure than the effect of the surface pressure bias bug in ERA5.P.

2.3 | Assimilation of IBTrACS observations

As detailed in Bell *et al.* (2021), ERA5.P had assimilated pressure reports from the IBTrACS (Knapp *et al.*, 2010) in a suboptimal way that led to the occasional severe overamplification of tropical cyclones analysed. The problem was not so much the quality of this rich dataset of bogus observations; it was the way that the information had been ingested in ERA5.P. In contrast to other in-situ observations, the first-guess check and variational quality control had been switched off to ensure that these sparse and valuable observations were not disregarded. In addition, the assigned observation error of 0.78 hPa was quite low for these bogus observations; that is, they were given a high (relative) weight in the assimilation, especially in unobserved regions where the estimate of the model first-guess quality is low. Although ERA5.P had fitted the data very well at observation locations, the assimilation system was prone to shifting the location of the storm centre away from the IBTrACS location, leading to too intense cyclones. As was shown in Bell *et al.* (2021, fig. 6), ERA5.P had a number of cases of minimum pressure well below 900 hPa and significant wave height well above 20 m. Extremes were up to 840 hPa and 34 m respectively, which is very unrealistic. Statistics were also very much out of line with the ERA5 period from 1979 onwards, where IBTrACS observations had not been assimilated, which can be seen from a comparison of fig. 6a,b in Bell *et al.* (2021).

For the ERA5 production from 1940 to 1978, the strategy was revised to make the assimilation of IBTrACS safer. First of all, the observation error assigned to these

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observations was increased from 0.78 to 2.0 hPa for streams 5–8 and to 2.5 hPa for streams 1–4. This is more aligned with the expected accuracy of these observations and is similar to what was used in the Coupled ECMWF Reanalysis of the 20th Century (Laloyaux *et al.*, 2018) and Twentieth Century Reanalysis (20CR) version 3 (Slivinski *et al.*, 2019) reanalyses. The larger value for streams 1–4 reflects the lower quality expected for the oldest observations and was also introduced to ensure that the very extreme cases did not reappear for the earliest two decades, since this was the era (at least for the 1950s) when the excessive extremes were found to be most serious in ERA5.P (not shown).

Secondly, the Huber norm method of VarQC (Tavolato & Isaksen, 2015) was reintroduced. This mechanism that is in place for other in-situ observations lowers the weight given to the observation so it has a smaller effect on the analysis when it differs significantly from the ERA5 first-guess estimate. The associated first-guess check was also reintroduced. It should be noted that this check is set at quite a relaxed level. Only when the magnitude of the departure from the model first guess exceeds 18 times the quadrature addition of the background and observation errors is the observation disregarded altogether. Otherwise, the observation is assimilated, although with much reduced weight for the larger departures. For the IBTrACS observations this ratio was increased to 25 for ERA5; that is, for a first-guess state with high confidence and observation error of 2.5 hPa the limit for rejection is 62.5 hPa. For less certain states this limit is even less strict. It only filters out observations that are absolutely out of sync with the ERA5 first guess. In addition, IBTrACS observations were capped at 910 hPa; that is, when a lower value was reported, this capped value was used rather than disregarding the observation. This cap was introduced because the limited resolution of the ERA5 product is not expected to be able to provide sensible results much deeper. In addition, it has been shown (Majumdar *et al.*, 2023) that the ECMWF model version (IFS Cy41r2) used by ERA5 likely overestimates surface drag for extreme cases. Such behaviour was also observed and addressed by Breivik *et al.* (2022) in a study for an area around northwest Europe that used the ocean-wave model (WAM), a version of which is also used in ERA5. As a result, in ERA5, the correct extreme values and structure of surface pressure and wind can lead to a (severe) overestimation of ocean wave height. Therefore, for some such cases, ERA5 cannot get surface pressure and significant wave height both right at the same moment. It is essential that wave height is not too excessive; that is, the surface pressure in ERA5 has to give. As described in Majumdar *et al.* (2023), this has now improved in the latest ECMWF model cycle, and it is expected that the next generation of ECMWF reanalysis (ERA6) will benefit from that.

For the ERA5.P period, the effect on data usage is demonstrated in Figure 5. It shows that ERA5 only removes slightly more observations. Of course, the weight given to each observation is much reduced, especially for large first-guess departures where VarQC is active. The rejections in ERA5.P occur for a large part over Madagascar and the southeast of mainland Asia. For ERA5, there are additional rejections over sea.

Figure 6a shows a density plot between IBTrACS observed pressure (original values; i.e., without the capping at 910 hPa) and ERA5 analysed pressure at the cyclone centre for the full back extension period from 1940

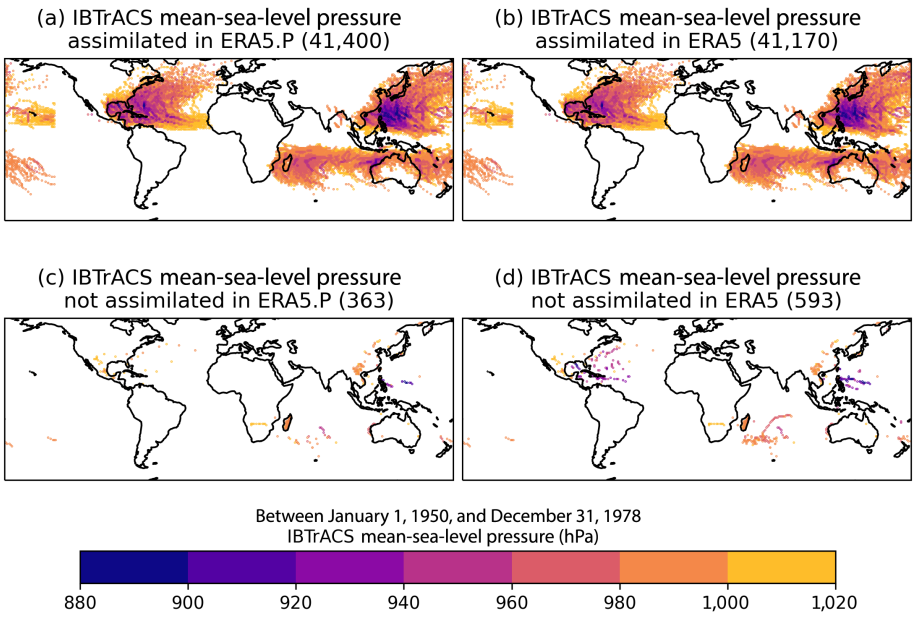

FIGURE 5 Maps of IBTrACS mean-sea-level pressure data assimilated in (a) ERA5.P and (b) ERA5. (c, d) Data that were not assimilated. Numbers in parentheses indicate the observation counts. ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; ERA5.P, ERA5 preliminary extension back to 1 January 1950; IBTrACS, International Best Track Archive for Climate Stewardship.

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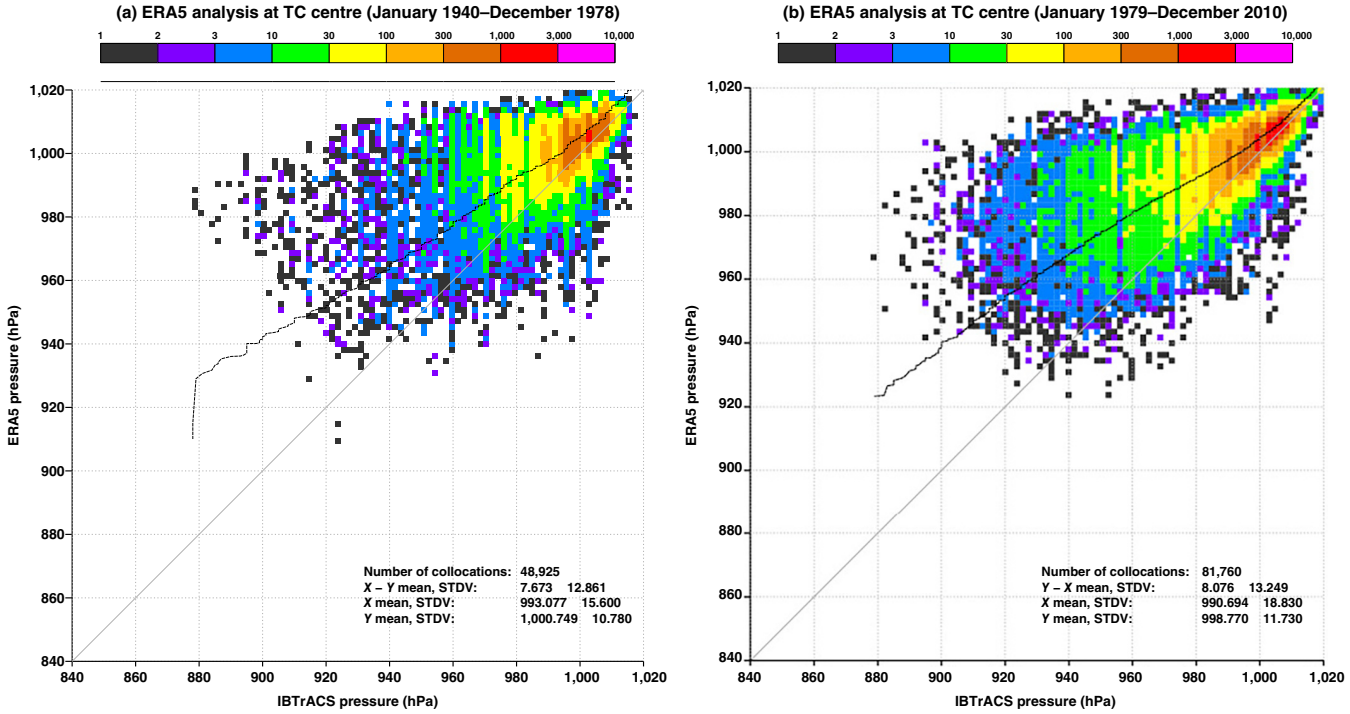

FIGURE 6 Density plot for the comparison between IBTrACS observations and the closest minimum of ERA5 pressure for (a) the 48,925 observations that were available between 1940 and 1978 and (b) the 81,760 observations between 1979 and 2010. Panel (b) is a copy of fig. 6b from Bell *et al.* (2021). For panel (a) the mean IBTrACS and ERA5 values are 993.1 and 1000.7 hPa respectively, and the STDV between the two is 12.9 hPa. For panel (b) these numbers are 990.7, 998.8, and 13.2 hPa respectively. The thin black line is the cumulative distribution function match between both quantities (smallest versus smallest, second smallest versus second smallest, etc.). ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; IBTrACS, International Best Track Archive for Climate Stewardship; STDV, standard deviation; TC, tropical cyclone.

to 1978. Figure 6b, which is a copy of fig. 6b from Bell *et al.* (2021), shows a similar density plot for the period from 1979 to 2010. Although now the strength of the extreme cases is usually underestimated, it looks much more in line with the ERA5 density plot for 1979 to 2010. In other words, the representation of tropical cyclones is now much more homogeneous over the entire 83-year ERA5 period.

The most extreme ERA5 case before 1979 occurred for Super Typhoon *Vera*, which was an exceptionally intense (Category 5) tropical cyclone that struck Japan in late September 1959. The two lowest squares in Figure 6 correspond to *Vera* for 0000 UTC and 0600 UTC 26 September 1959; that is, just prior to its landfall over Honshu. The ERA5 estimated minimum pressures are 915 and 910 hPa respectively, which are slightly deeper than the value of 923 hPa for both times in the IBTrACS database, and the difference in location is about 40 km (i.e., just over one ERA5 grid point). Therefore, ERA5 did overestimate this storm at that moment; maximum ERA5 wave height is 25 m, which is probably too intense. According to the IBTrACS observations, *Vera* had its deepest pressure of 895 hPa on 23 September 1959, at which time the ERA5 minimum pressure was 985 hPa (i.e., far too shallow).

The details of the evolution of *Vera* and the large scatter in Figure 6 show that it was still challenging to assimilate IBTrACS observations well at all occasions. However, as already said, ERA5 pre-1979 shows much more consistent results with ERA5 post-1978 than ERA5.P does. The consistency also applies to the extreme mean-sea-level pressure and wave climate globally, as illustrated in Figure 7, which shows grid-point-wise minimum pressure (Figure 7a,c,e) and maximum significant wave height (Figure 7b,d,f) for all values at 0000, 0600, 1200, and 1800 UTC over the periods indicated and product. ERA5.P shows much more intense low pressure (Figure 7a) and significant wave height (Figure 7b), despite the much shorter aggregation period (29 years vs. 42 years for the other panels). Minimum pressure and maximum significant wave height reach 841 hPa and 31 m respectively. The results for ERA5 are much more consistent between the first and last 42 years (note that the overlap of 1981 was included to make each period equally long). These extremes more or less highlight the same distribution of tropical cyclones across both eras. The maximum of 25 m

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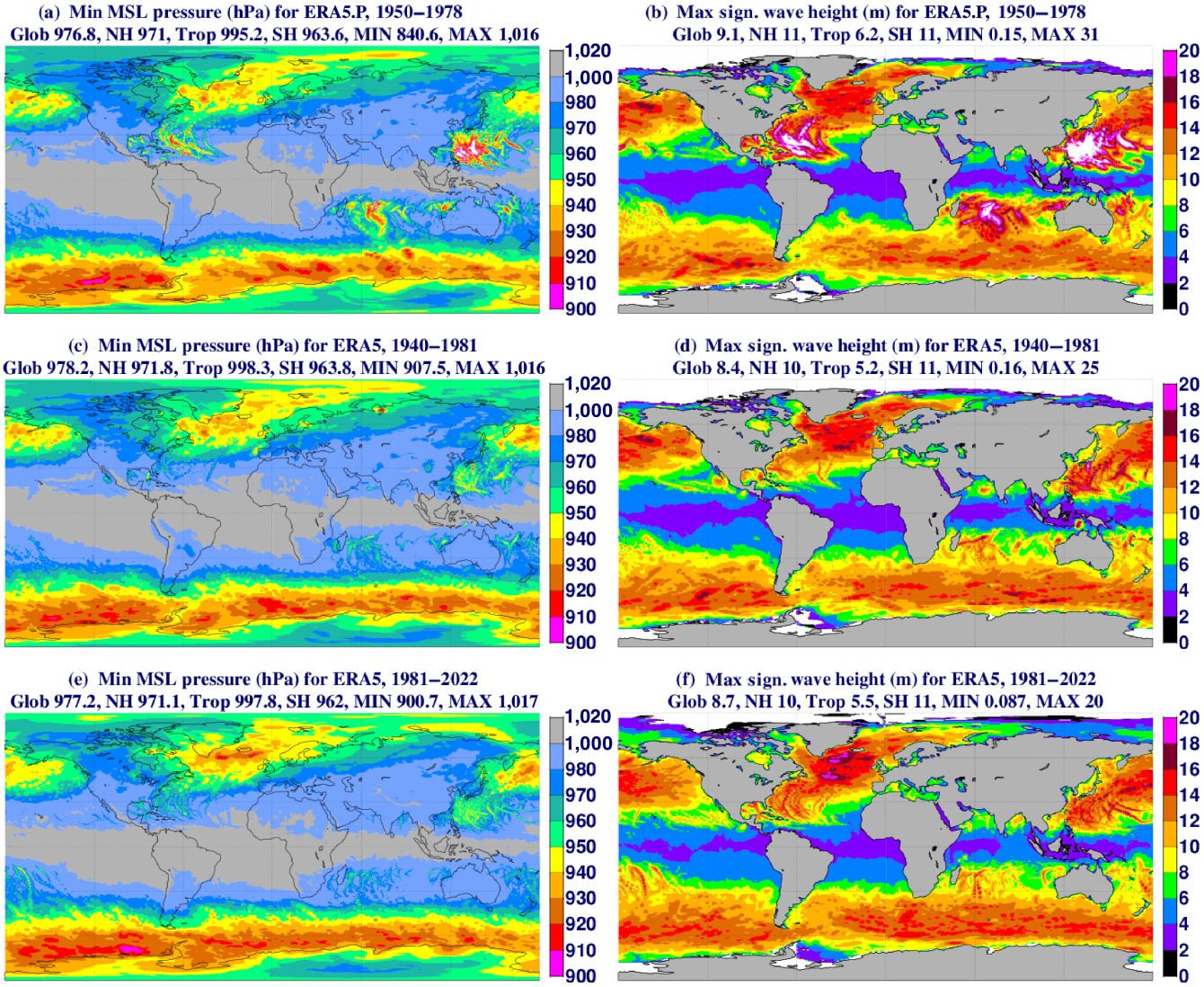

FIGURE 7 Grid-point-wise (a, c, e) minimum mean-sea-level pressure (hPa) and (b, d, f) maximum significant wave height (m) for six-hourly fields on the native grid for (a, b) ERA5.P from 1950 to 1978, and for ERA5 from (c, d) 1940 to 1981 and from (e, f) 1981 to 2022. Note that the year 1981 is included in (c)–(f) to make each indicated period exactly 42 years long. For each panel the area-weighted average of the field in question is provided over the globe (Glob), north of 20°N (NH), between 20°N and 20°S (Trop), and south of 20°S (SH) as well as the minimum (MIN) and maximum (MAX) values. ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; ERA5.P, ERA5 preliminary extension back to 1 January 1950; MSL, mean sea level.

wave height was obtained for *Vera* in 1959 as already detailed. On no other occasion did significant wave height exceed 20 m. It is interesting to see that local extreme values have intensified with time over the North Atlantic and Southern Ocean, which is likely related to climate change.

3 | EVOLUTION OF SYNOPTIC PERFORMANCE

As mentioned in Section 1, the assimilation of as many observations as possible inherently brings temporal improvements in a reanalysis, as the observing systems develop over time. This section presents two methods to quantify how these improvements manifest themselves in the ERA5 products and a case study that qualitatively assesses synoptic skill in the early 1940s. Both methods present distinct merits and limitations, and these are also discussed.

3.1 | Forecast skill

A first method, inherited from reanalysis's origins in numerical weather prediction, is to consider the quality of forecasts, measured against verifying analyses. Figure 8

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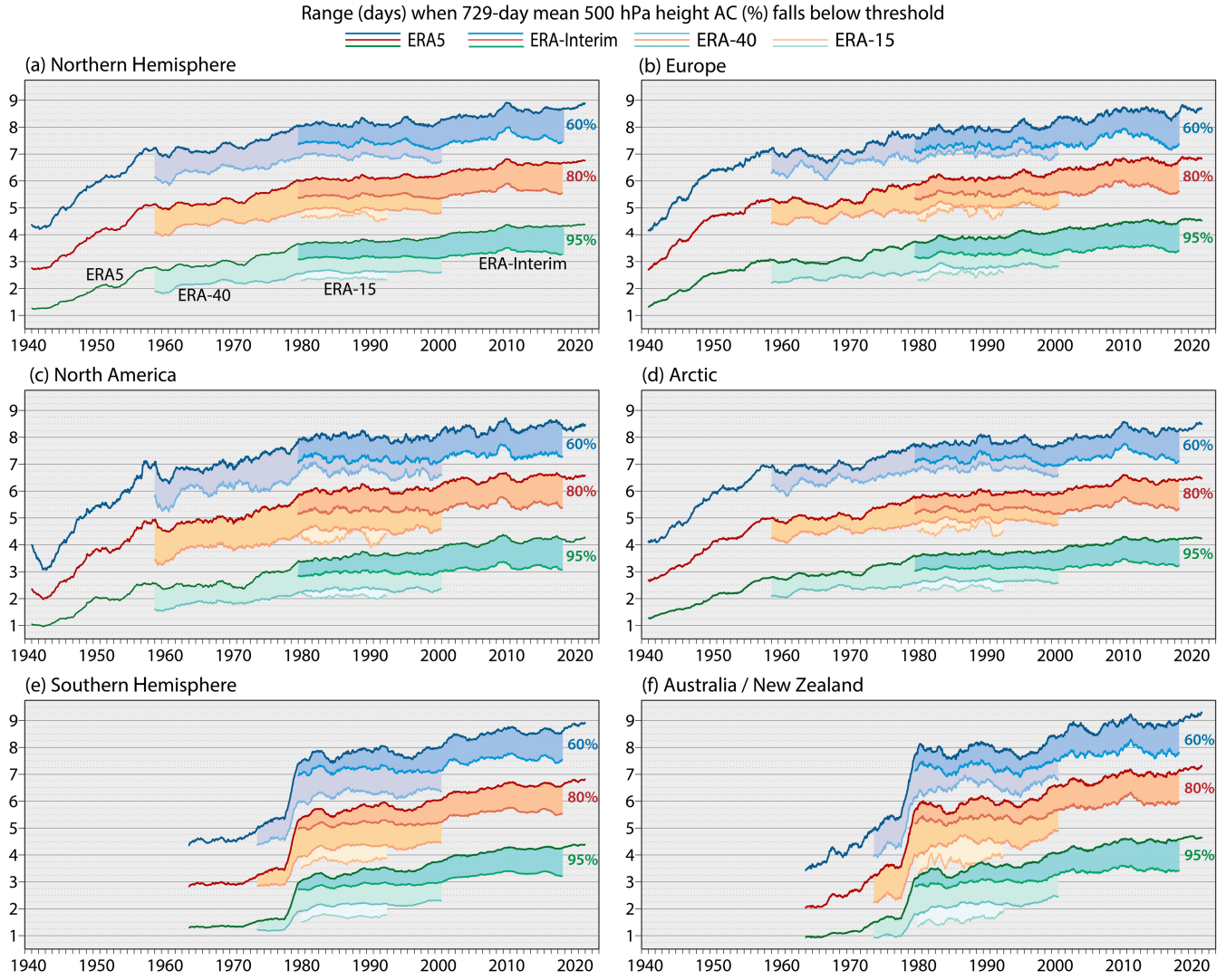

FIGURE 8 Range (days) at which running 729-day mean anomaly correlations of 500 hPa height forecasts at 0000 and 1200 UTC from 1940 to 2022 reach 95% (green), 80% (orange), and 60% (blue) for (a) Northern Hemisphere, (b) Europe, (c) North America, (d) Arctic, (e) Southern Hemisphere, and (f) Australia/New Zealand. The heavy lines denote ERA5, and the thin lines denote ERA-Interim, ERA-40, and ERA-15. Shading denotes the difference between ERA5 and other reanalysis products during the period for which both are available. ERA: European Centre of Medium-range Weather Forecasts Reanalysis (-5: v.5; -Interim: January 1979–August 2019; -40: September 1957–August 2002; -15: 1979–1993).

shows the temporal evolution of the mean anomaly correlation coefficient (ACC) of 500 hPa geopotential height forecasts for six regions. As in Bell *et al.* (2021), results are shown only for the periods for which the availability of weather observations gives confidence in the use of ERA5 analyses to verify the forecasts. Overall, in all panels, the plots show an upward trend of the forecast skill that can be attributed to the improvements in the observing system. This is particularly noticeable over the Southern Hemisphere, where the marked increase in skill at the end of 1970s can be directly linked to the improved quality of reanalyses following the assimilation of TIROS operational vertical sounder data. At the same time, for the Northern Hemisphere the improvement is not so evident because on the one hand it is better observed than the Southern Hemisphere and on the other hand the satellite data are an important component of the observing system over the oceans which have a greater extent in the Southern than the Northern Hemisphere. The plots also illustrate the incremental improvement of the global reanalyses produced at ECMWF as a result of the progress in the developments of data assimilation methods and the forecast model. An example is yet again the Southern Hemisphere over which the ACC of 95% for ERA5 reaches day 1 a decade earlier than for ERA-40. The poor forecast skill over the Southern compared with the Northern Hemisphere in the 1960s, when the ACC of 95% barely reaches day 1, followed by a step change in performance with the

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advent of TIROS operational vertical sounder data, is an indication that the future generations of global reanalyses can further be improved provided that additional historical observations can be made available.

3.2 | Ensemble spread

A second method to evaluate the synoptic quality of the ERA5 analyses is to consider the dispersion between the ERA5 members in the ERA5 ensemble. The ensemble spread is dictated by the amount of observations and by the uncertainty hypotheses imposed on the data assimilation system. Under optimal conditions of a well-observed system and a well-calibrated data assimilation system, a small (large) ensemble spread would be indicative of high (low) confidence. Fewer observations generally lead to higher ensemble spread.

It has been noted that the ERA5 ensemble is generally underdispersive (i.e., overconfident)—for example, Rodwell, personal communication (2024) based on a methodology by Rodwell *et al.* (2018). For this reason, the ensemble spread should be used in a qualitative manner, such as "the larger spread in period A compared with period B indicates a larger uncertainty for period A", rather than quantitatively like "the ERA5 uncertainty of surface temperature at this location and time is 0.2 K."

Given this caveat, the variations in ensemble spread are now discussed. For several upper air parameters the evolution in ensemble spread was demonstrated in Hersbach *et al.* (2020, fig. 12) for the satellite period from 1979 onwards. For the extension back to 1940 it is clear that the lack of satellite observations before the 1970s will have a large impact, in line with the coverage changes discussed in Section 1 and the change in forecast skill of Figure 8.

This is illustrated by Figure 9, which shows maps of the average ERA5 ensemble spread for surface pressure (Figure 9a,c,e,g,i) and 2-m temperature (Figure 9b,d,f,h,j) over each 3 h in the 1940s, 1950s, 1980s, and from 2013 to 2022 on a 0.5° regular latitude–longitude grid. Figure 9a,b shows the 83-year average ERA5 synoptic variability (0.25° regular latitude–longitude), which resembles a system without any synoptic skill. They are the average of the STDV for subsets over all three-hourly values between 1940 and 2022, each for one particular month and hour of the day (i.e., the average of 12×8 separate statistics) such that variability of the daily and seasonal cycle is suppressed. Similar statistics from the ERA5 EDA control show, apart from the lower resolution, near identical results (not shown). Plots restricted to 1940–1979 (pre-satellite), 1980–2019 (satellite) and 1991–2020 (official World Meteorological Organization climate period) are all very similar as well (also not shown). From Figure 9a,b it is evident that the natural variability, as estimated by ERA5, varies highly over the globe, with largest values for pressure over areas poleward from the mid-latitude storm tracks and for temperature at high northern latitudes over land and near the coast of Antarctica (i.e., variability is high even without seasonal variability). Over ocean the temperature varies much less than over land.

For both parameters, the 1940s ensemble spread is not far from the ERA5 variability over a large part of the Southern Hemisphere, indicating that ERA5 is not able to resolve synoptic situations well over these regions for this time period. The situation is much better over the Northern Hemisphere, especially over the United States and large parts of Europe where ensemble spread is about one magnitude below variability. So, over these regions the estimation of the synoptic situation should, in general, be reasonably trustworthy. Examples are provided in Sections 3.3 and 5.5.

For the 1950s the situation is already much improved, especially for surface pressure. Ensemble spread is still very large over the Southern Ocean, but it is well below the climatological variability. For 2-m temperature the improvement is also evident in some places (like Antarctica), though not for many others, such as over large parts of the Southern Ocean, South America, and South Africa. As expected, for early decades, areas of large ensemble spread correlate well with observation coverage maps as presented in Figure 1. For later decades, ensemble spread is reduced significantly, in particular from the 1980s when satellite observations became available in large quantities (Figure 2). Since then the ensemble spread has continued to reduce over the Southern Hemisphere in particular, with hardly any visible difference left between both hemispheres for recent years. Overall, the ensemble spread is smallest over Europe for the latest 10-year period, which corresponds to the location of highest in-situ observation density as indicated in the right-hand panels of Figure 1.

3.3 | A case study for February 1941

An example of synoptic skill in the early ERA5 period over Europe is an extreme storm that swept over the Iberian peninsula in February 1941 and led to severe damage and disruption over Portugal and northwest Spain (Muir-Wood, 2011). Its hourly representation by ERA5 is shown in Figure 10a–c for three moments in its evolution and ERA5 ensemble spread in Figure 10d–f. Although the number of observations that were available is low (for the plotted area indicated by circles), mainly surface

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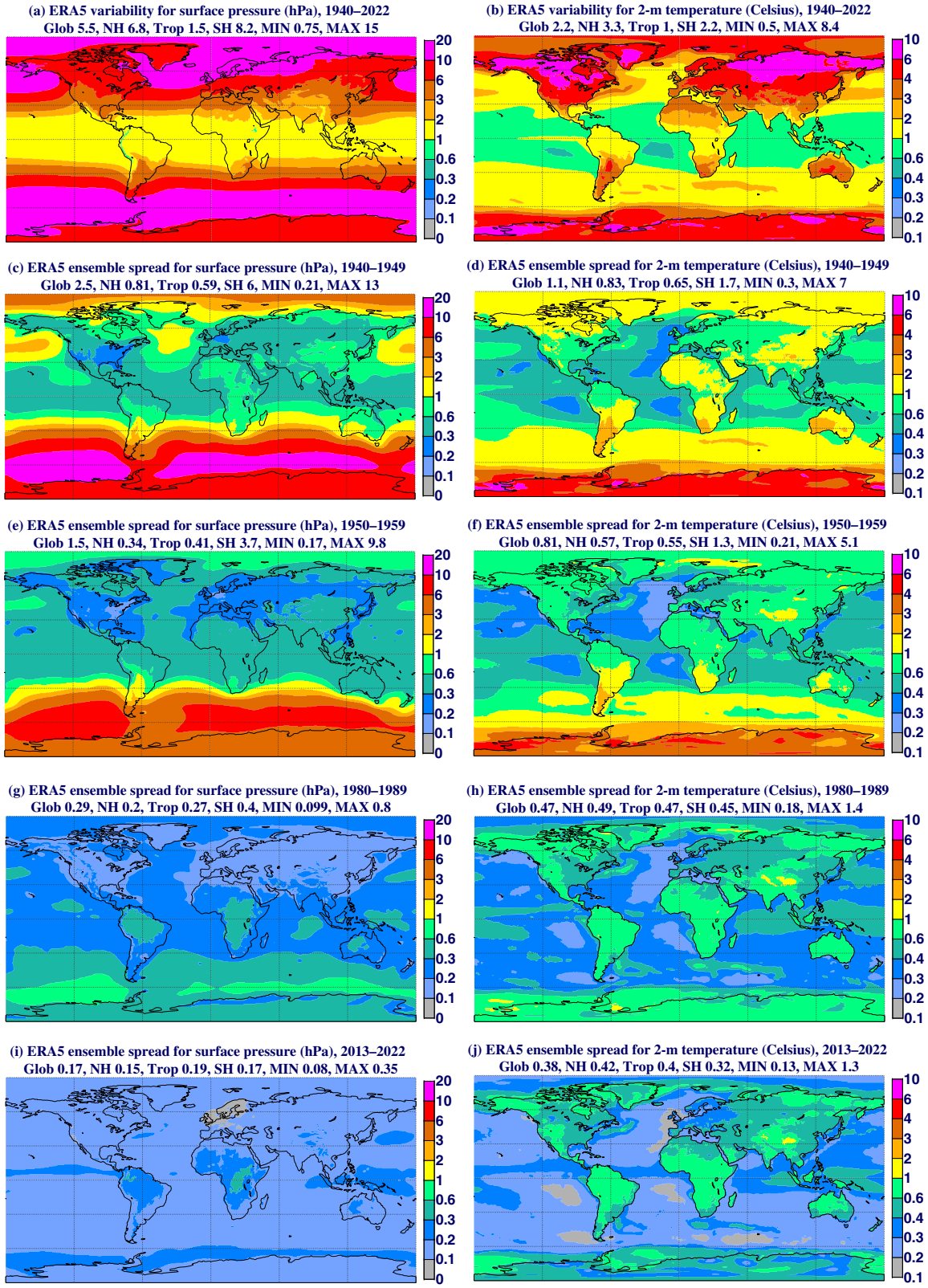

FIGURE 9 European Centre of Medium-range Weather Forecasts Reanalysis v.5 (ERA5) synoptic variability over 83 years excluding the daily and seasonal cycle (a, b) and 10-year mean ERA5 ensemble spread for (c, d) 1940–1959, (e, f) 1960–1969, (g, h) 1980–1989, (i, j) 2013-2022, for (a, c, e, g, i) surface pressure and (b, d, f, h, j) 2-m temperature. For each panel the area-weighted average of the field in question is provided over the globe (Glob), north of 20°N (NH), between 20°N and 20°S (Trop), and south of 20°S (SH) as well as the minimum (MIN) and maximum (MAX) values.

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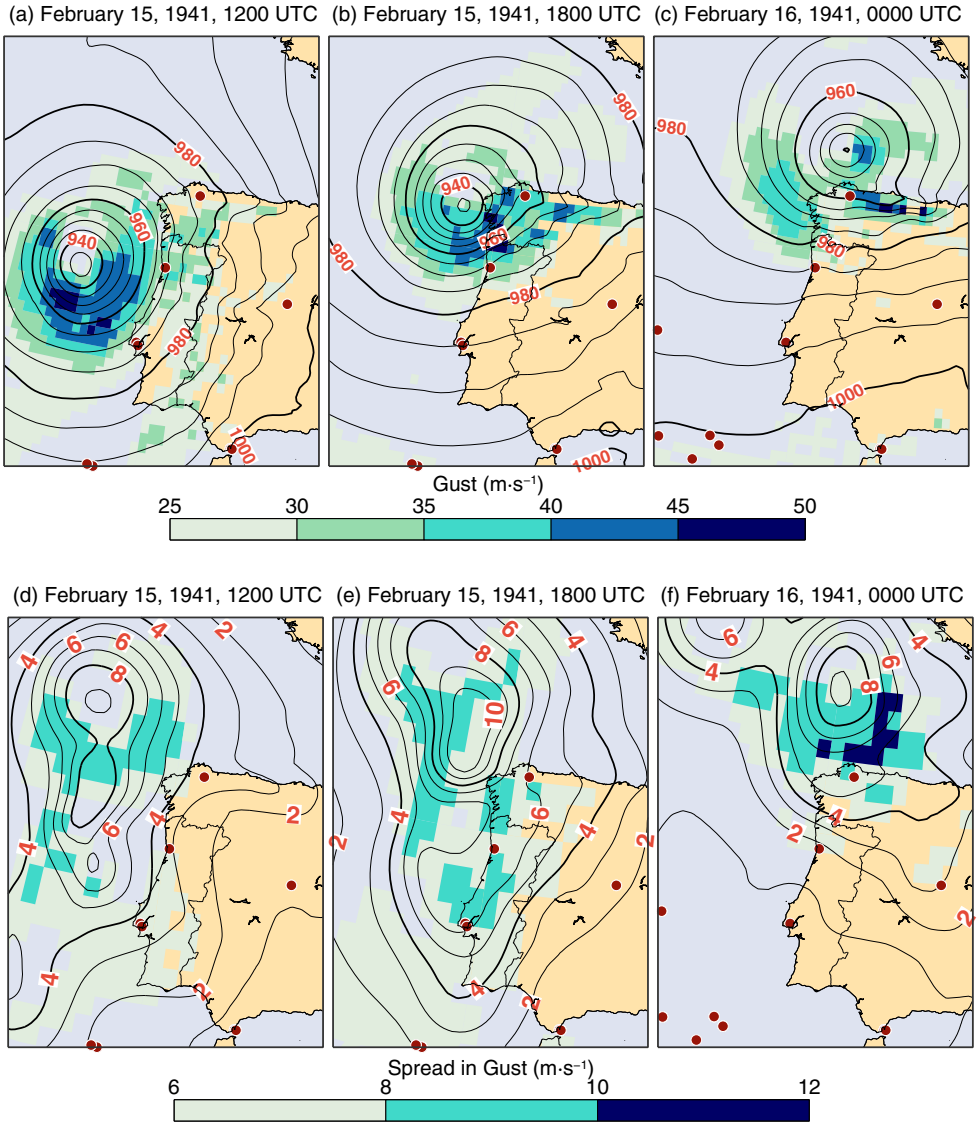

FIGURE 10 Three moments of the Iberian storm of 1941 for (a–c) European Centre of Medium-range Weather Forecasts Reanalysis v.5 (ERA5) and (d–f) ERA5 ensemble spread for mean-sea-level pressure (contours, hPa) and hourly maximum 10-m gusts (colours, $m \cdot s^{-1}$). The locations of assimilated pressure and marine-wind observations from which these ERA5 reanalysis fields were constructed are shown as red circles.

and mean-sea-level pressure plus surface marine wind, the ERA5 analysis does show an extraordinary extreme situation. Maximum gust of around 45 $m \cdot s^{-1}$ is in line with reports that are mentioned in Muir-Wood (2011), with the caveat raised by the authors that these originated from recorders that were not necessarily calibrated to such extreme conditions. Minimum pressure (around 940 hPa) is below an analysis from map 2 from Muir-Wood (2011) by P. Mateo González, which places the storm track closer to the coast than ERA5 does. A study of this storm case for the 20CR reanalysis (Compo *et al.*, 2011) by Baumann and Reichen (2017) also indicates a less deep minimum pressure. Therefore, the storm was extreme in both ERA5 and 20CR. However, the lack of (particularly marine) observations does not allow for clarity whether the track as suggested by ERA5 was located more over sea and even more intense than previously believed. The large ERA5 ensemble spread as presented in Figure 10d–f confirms the locally large uncertainty. Interestingly, the largest spread (in both mean-sea-level pressure and maximum gust) is somewhat located north of the minimum pressure, indicating a larger uncertainty in position in the north–south direction than in the east–west direction.

4 | CHARACTERISTICS OF HYDROLOGICAL VARIABLES FROM 1940 TO 2022

4.1 | Soil moisture and precipitation

Bell *et al.* (2021) identified the initialisation of soil moisture as one of the deficiencies of ERA5.P. Its production streams had generally been started with soil water contents that were much lower than those typical of post-1978 production. The response was a moistening (or spin-up) of

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each production stream that was generally slower for the deeper layers, on a time-scale that varied from continent to continent. Sharp changes in soil moisture occurred when one stream ended and another began, something that also happened in post-1978 production at the beginning of 2015.

Figure 11 presents volumetric water content anomalies averaged over continental regions for each of the four modelled soil layers of ERA5 and for the deepest layer of ERA5.P. It shows that although low pre-1979 values of the deepest layer soil moisture remain an issue for ERA5, the spin-up of soil moisture has been reduced appreciably in ERA5 by starting the production streams either from the scout reanalysis that began in 1939 or from ERA5.P values that were partially spun up. A smaller improvement is found for the second deepest level (not shown). It must be noted here that the corresponding figure in Bell *et al.* (2021, fig. 27), has been found to be incorrect for all but the deepest soil layer, due to a programming error that resulted in some wrong data being plotted. The older figure is substantially wrong for Africa and Australia, and the change from ERA5.P to ERA5 accordingly cannot be judged for the top three layers by comparing the new and old figures.

It was suspected that several instances of relatively high surface air temperature and low precipitation that were found in ERA5.P for the 1950s and 1960s, but not in independent datasets, were consequences of persistently low deep-soil moisture, exacerbated by gaps in the availability of observations whose assimilation would have compensated (Bell *et al.*, 2021; Simmons *et al.*, 2021). This view has been supported by a substantial reduction in the severity of these instances in ERA5. Figure 12 provides an example. It shows temperature and precipitation anomalies over Africa for 1965–1966, a period when there was a dearth of available observations of 2-m temperature and humidity over large parts of the continent. Reanalysis results are shown from the recently released JRA-3Q product (Kobayashi *et al.*, 2021; Kosaka *et al.*, 2024) as well as from ERA5 and ERA5.P.

In this case ERA5.P has a zone extending from Nigeria to Ethiopia where temperatures are 1–4°C higher than the 1991–2020 average, in stark contrast to the below-average

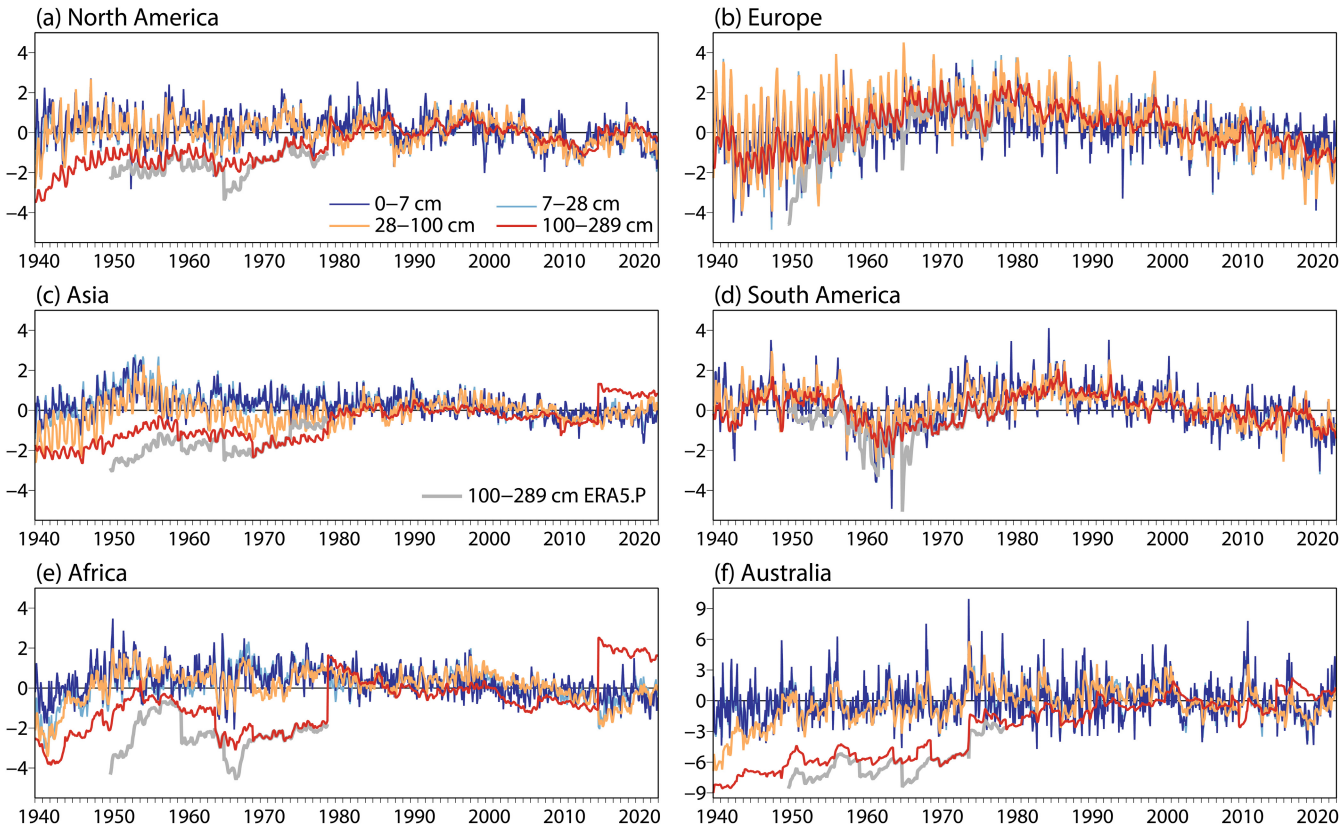

FIGURE 11 Monthly mean volumetric soil moisture anomalies (%) relative to 1991–2020 from ERA5 for the four layers of its land-surface model, averaged over land areas designated to be (a) Europe (land bounded by 25°W–40°E and 34°N–72°N), (b) North America (170°W–50°W, 15°N–85°N), (c) Africa (25°W–55°E, 40°S–35°N), (d) South America (90°W–25°W, 65°S–15°N), (e) Asia (60°E–180°E, 0°–85°N) and (f) Australia (110°E–160°E, 50°S–10°S), from 1940 to 2022. The depths of the layers are shown in the legend. Values for the lowest layer are shown for ERA5.P (grey) and ERA5 (red). ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; ERA5.P, ERA5 preliminary extension back to 1 January 1950.

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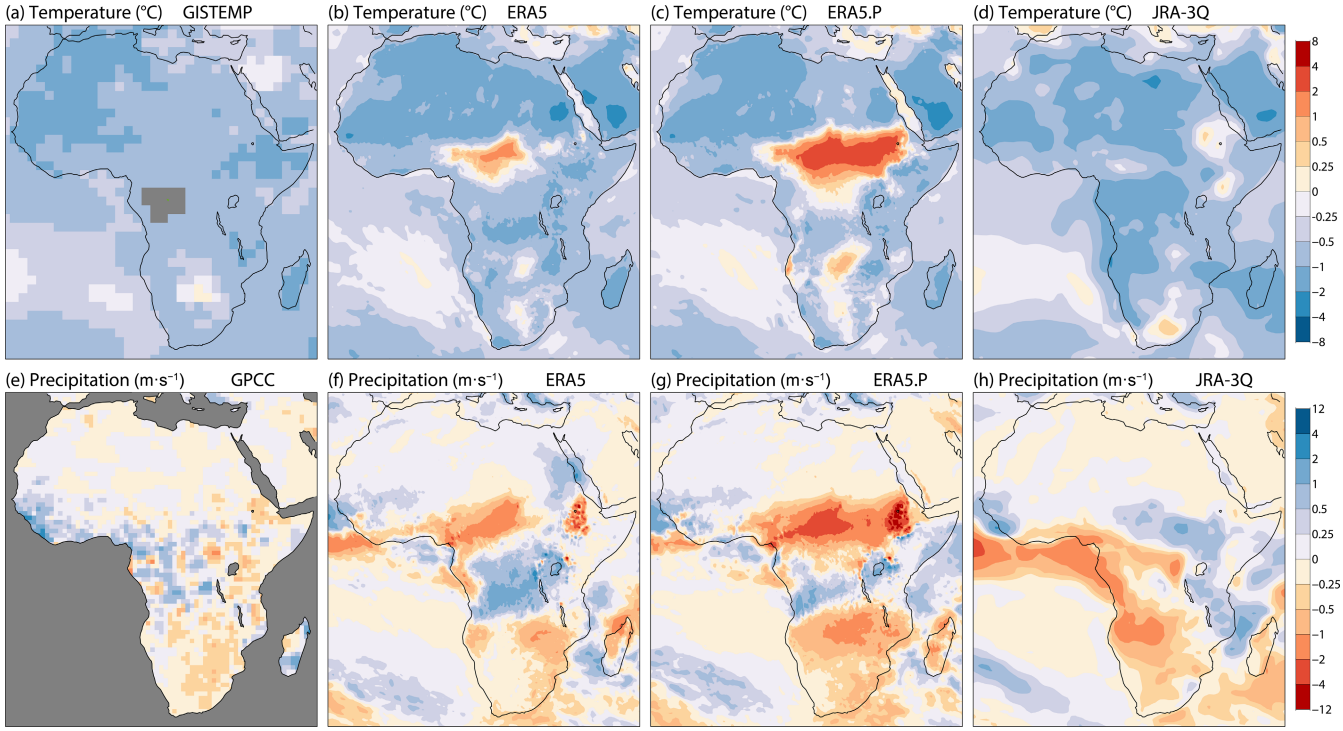

FIGURE 12 Anomalies in (a)–(d) surface temperature (°C) and (e)–(h) precipitation rate (mm·day^{-1}) for 1965–1966 relative to 1991–2020 from (a) GISTEMP and (e) GPCC, (b, f) ERA5, (c, g) ERA5.P, and (d, h) JRA-3Q. ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; ERA5.P, ERA5 preliminary extension back to 1 January 1950; GISTEMP, Goddard Institute for Space Studies Surface Temperature Analysis; GPCC, Global Precipitation Climatology Centre; JRA-3Q, Japanese Reanalysis for Three Quarters of a Century.

temperatures for the region shown by the Goddard Institute for Space Studies Surface Temperature Analysis (GISTEMP) dataset (Lenssen *et al.*, 2019) and by the Japanese Reanalysis for Three Quarters of a Century (JRA-3Q). Precipitation anomalies of several millimetres per day over the zone from ERA5.P are supported neither by the values from the Global Precipitation Climatology Centre (GPCC; Becker *et al.*, 2013)—here the 2022 version of its Full Data Monthly Product—nor by JRA-3Q. The temperature and precipitation anomalies are both still present in ERA5, but both are much weaker than in ERA5.P.

Agreement among datasets is better over other parts of the continent, in particular regarding the relatively cold temperatures in the northwest, above-average precipitation over the western Sahel, and dry conditions in the south.

Two other examples were presented by Simmons *et al.* (2021) for ERA5.P, over South America in 1961 and Africa in 1950–1952. In the South American case, anomalously high temperatures over the Amazon Basin have been reduced in ERA5. Anomalously high temperatures over the Congo Basin in 1950–1952 in ERA5.P have been replaced in ERA5 by temperatures lower than the 1991–2020 average. In both cases ERA5 is the closer of the two ERA reanalyses to GISTEMP and JRA-3Q. The same is true of the comparison of precipitation with GPCC and JRA-3Q.

Figure 13 shows 12-month running-mean rates of precipitation for six continental regions and from 1940 to 2022, for ERA5, ERA5.P, JRA-3Q, and GPCC. ERA5.P is close to ERA5 for most of its 1950–1978 period, but occasional deviations are evident, including the 1950–1952 and 1965–1966 spells over Africa and the 1961 spell over South America noted earlier.

Discrepancies in the overall amounts of precipitation between the reanalyses and GPCC are not of immediate concern because of GPCC's acknowledged provision of data that have not been adjusted for measurement biases, an issue discussed earlier in assessing precipitation from ERA-Interim Simmons *et al.*, 2010. Of more concern are the long-term shifts in precipitation rate of ERA5 relative to GPCC. These are essentially as discussed already by Bell *et al.* (2021) for the period 1950–2020 based on ERA5.P up to 1978. A point of note was that ERA5 did not reproduce GPCC's unusually high average precipitation rate over Asia for 2020. It now can be seen in Figure 13 that the same holds for 2021 and 2022, although to a lesser extent. The three years with highest Asian rainfall are 2020, 2021,

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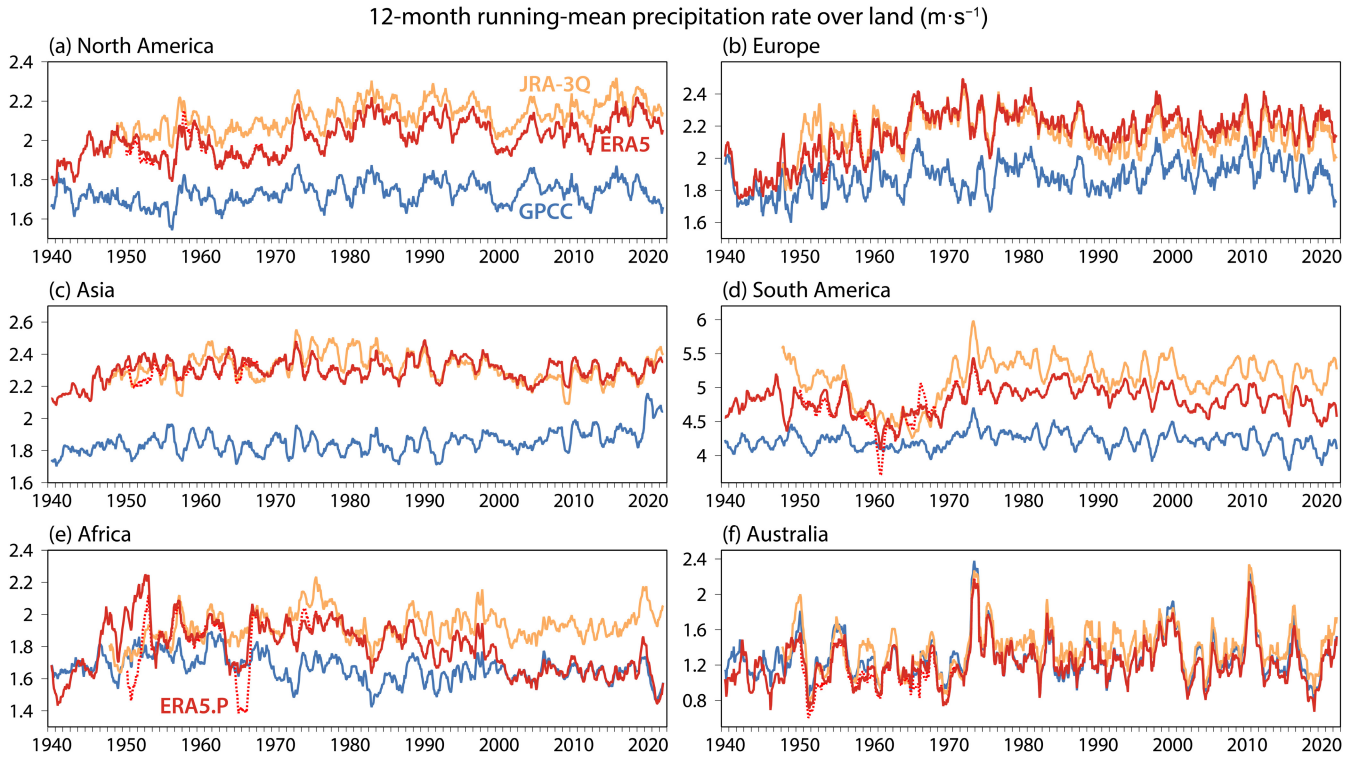

FIGURE 13 The 12-month running-mean precipitation rates (mm·day^{-1}) for ERA5 (red, solid), ERA5.P (red, dotted), JRA-3Q (orange), and GPCC (blue) from 1940 to 2022 over the continental land areas defined in the caption of Figure 11. The 2022 version of the GPCC Full Data Product is used for 1940–2020, and the corresponding Monthly Monitoring Product for 2021 and 2022. The JRA-3Q time series are from September 1947 onwards. ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; ERA5.P, ERA5 preliminary extension back to 1 January 1950; GPCC, Global Precipitation Climatology Centre; JRA-3Q, Japanese Reanalysis for Three Quarters of a Century.

and 2022 for GPCC. This is not the case for ERA5 and JRA-3Q. Reasons for this discrepancy are not yet clear.

ERA5 and JRA-3Q precipitation rates are generally similar in their overall levels and their interannual and longer term variations, throughout their common period from September 1947 onwards. Agreement is for the most part better than the agreement between ERA5.P/ERA5 and the earlier Japanese 55-year Reanalysis Project (JRA-55; Kobayashi *et al.*, 2015), for which corresponding comparisons are presented in Bell *et al.* (2021, fig. 21). The most disconcerting discrepancy between ERA5 and JRA-3Q concerns African rainfall since the turn of the century. JRA-3Q, like GPCC, shows little indication of the reduction in rainfall that Hersbach *et al.* (2020) reported for ERA5.

ERA5 precipitation rates generally increase over time during the 1940s relative to GPCC. This is likely due in part to the initial specification of anomalously small values of deep-soil moisture illustrated in Figure 11. A second factor is a bias towards low values of tropospheric temperature prior to the assimilation of sufficient radiosonde data. This temperature bias is illustrated later (in Figure 18) for global averages, at 50 hPa and at 500 hPa. The relatively low tropospheric temperatures in early years are accompanied by relatively low values of total column water vapour (not shown). Precipitation is initially relatively low over land but high over sea, where surface air temperature is constrained by the prescribed SST.

Time series of the correlations between monthly mean ERA5 and GPCC precipitation rates are shown in Figure 14 for the six continental regions. Correlations are initially much higher for Europe and North America than for other regions. They generally increase over all regions from 1940 to 1970, but with poorer performance over some regions in the mid 1960s, likely associated with the low number of surface observations at the time. Correlations change little over much of the subsequent period, but they tend to decline in recent years by amounts that vary regionally. This is not necessarily due to a decline in quality of ERA5, as the GPCC products are based on fewer observations for recent years. Changes over time in the distribution of precipitation may also make analysis of its trends more difficult.

Correlations between GPCC and ERA5 and between GPCC and ERA5.P have been computed separately for the period 1950–1969. They show welcome increases from

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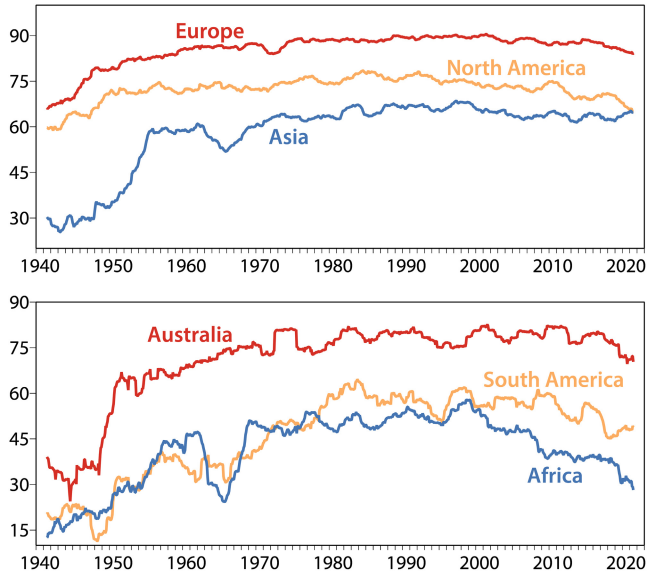

FIGURE 14 Correlations (%) between European Centre of Medium-range Weather Forecasts Reanalysis v.5 and Global Precipitation Climatology Centre monthly precipitation rates at 1° × 1° resolution for running 36-month periods from 1940 to 2022, over the continental land areas defined in the caption of Figure 11. Mean annual cycles for 1940–2022 are subtracted from each set of data prior to calculating the correlations.

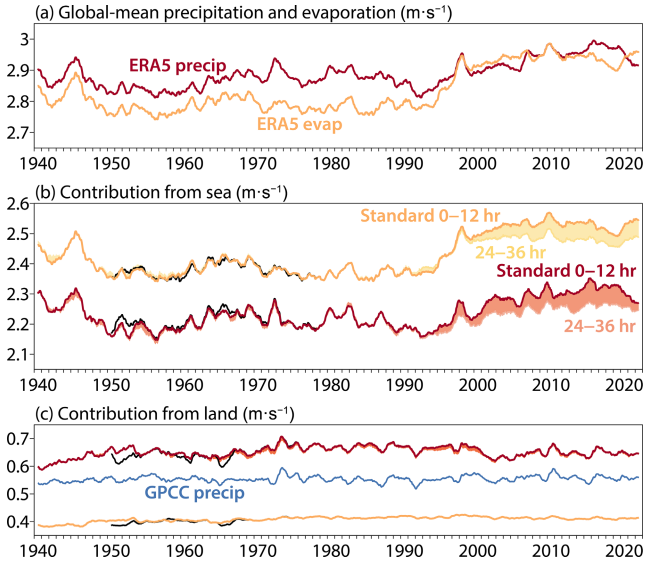

FIGURE 15 (a) The 12-month running averages from 1940 to 2022 of global-mean rates of precipitation (red) and evaporation (orange) from ERA5 (mm·day^{-1}). The contributions to these global averages from values over (b) sea and (c) land are shown for ERA5 and ERA5.P (thin black lines). They are accumulated from 24–36 h forecasts as well as the standard 0–12 h forecasts in the case of ERA5, with differences denoted by shading. Averages of precipitation over land (other than Antarctica) from GPCC are also shown (blue). ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; ERA5.P, ERA5 preliminary extension back to 1 January 1950; GPCC, Global Precipitation Climatology Centre.

ERA5.P to ERA5 from 30% to 36% for Africa and from 33% to 36% for South America. The two regions with the highest correlations are Europe (84%) and North America (73%), with insignificant differences between the ERA versions for these two regions.

4.2 | Global hydrological balance and spin-up

Figure 15 presents time series for 1940–2022 of global averages of evaporation and precipitation, and the contributions to these averages from sea and land, extending results shown for 1979–2019 by Hersbach *et al.* (2020). Amongst much else, it shows that the increase in precipitation over land noted above for the 1940s is not accompanied by a commensurate increase in evaporation. The corresponding decrease in precipitation over sea matches more closely the decrease in evaporation. It is larger than the increase over land when the two are expressed as contributions to the global average, but smaller when they are expressed as values per unit area. Sea and land contributions are also shown for ERA5.P: the occasional spells of smaller precipitation over land from ERA5.P are largely compensated by increased precipitation over sea. Differences between ERA5.P and ERA5 are smaller for evaporation than for precipitation.

Precipitation and evaporation are not quite in global balance for the 1950s and 1960s, but the imbalance is fairly uniform over these decades. Interannual variability in precipitation over land is lower then than in later decades, for both ERA5 and GPCC. Both the degree of imbalance and the ERA5 precipitation over land increase around 1970. Imbalance subsequently reduces in the late 1980s and early 1990s, but it is not until around the year 2000 that ERA5 precipitation reduces, returning the typical difference between ERA5 and GPCC precipitation to a value similar to that of the 1950s and 1960s. As reported by Hersbach *et al.* (2020), precipitation and evaporation are in quite good global balance for about 20 years from the mid 1990s. Imbalance then worsened, with a decline in evaporation that was not matched by a decline in precipitation from 2015. This was short-lived, however: the extended time series presented here show a reversal over the final 3 years.

Precipitation and evaporation increase over sea from the early 1990s to around 2010. The increase occurs in the Tropics (Bell *et al.*, 2021) and is of questionable realism (Buschow, 2023; Hersbach *et al.*, 2020). One indication of the latter is the sensitivity of results to the various ways

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in which precipitation and evaporation can be derived. The standard ERA5 values discussed in this article are accumulated during the first 12 h of the 18-h background forecasts run from 0600 and 1800 UTC. These are the values most readily accessible to users via the C3S Climate Data Store and for which monthly means are made available. As noted but not illustrated by Bell *et al.* (2021), precipitation rates over sea accumulated instead from the 24- to 36-h range of forecasts run from 0000 and 1200 UTC do not show as large a rise in the 1990s and 2000s. The differences between the standard values and 24–36 h forecasts, for evaporation as well as precipitation, are shown by shading in Figure 15b,c. Differences are small for much of the reanalysis period, but evaporation and precipitation over sea (but not land) are evidently lower for the 24–36 h forecasts than for the standard 0–12 h forecasts in the final two to three decades of ERA5.

The implied adjustment (or spin-up) of the hydrological cycle at short forecast ranges indicates that the observations assimilated for the recent data-rich years are in conflict with the assimilating model's preferred state. The degree of balance between global-average precipitation and evaporation accordingly depends on how precipitation and evaporation are defined. Table 2 presents the contributions from oceanic regions to global-mean evaporation and precipitation averaged from 2003 to 2022 for six different ways of accumulating values over successive 12-h periods.

The most basic representation of precipitation is that accumulated from 0900 to 2100 UTC and from 2100 to 0900 UTC from the 12-h "final trajectory" forecasts of ERA5's 4D-Var data assimilation (Hersbach *et al.*, 2020). The 4D-Var adjusts model states at 0900 and 2100 UTC such that the subsequent 12-h forecasts (known also as trajectories) fit observations optimally. This is done iteratively, with the final trajectory providing the hourly analyses, from which 18-h and 10-day forecasts are each run twice per day. The hydrological cycle is thus spun up during a segment of the final trajectory as well as during the forecasts themselves, as indicated by the effective spin-up periods shown in Table 2.

The final trajectory provides the lowest estimate of precipitation over sea. Marine precipitation increases as the effective spin-up period increases out to around 2 h ahead, but it decreases at longer range. Evaporation changes little during the first 24 h of spin-up, but it decreases at longer range when precipitation decreases.

Figure 16 presents sample maps of the changes in precipitation. It contrasts the low spin-up in averages for 1941–1970 (Figure 16a,c) with the high spin up for 2003–2022 (Figure 16b,d). The two phases of the spin-up are illustrated by the differences between standard and final-trajectory values (Figure 16a,b) and between 24–36 h forecasts and standard values (Figure 16c,d). Increases in precipitation during the first 24 h or so are widespread in the Tropics and Subtropics for 2003–2022, with decreases mainly over the islands of the Maritime Continent and a few other coastal areas. The pattern shifts markedly further into the forecast range, when changes over sea are more confined to the Tropics and dominated by bands of reduced precipitation in the intertropical convergence zones, with some continuing increase in precipitation in neighbouring bands located mainly to the north. Precipitation continues to increase over the Congo Basin, but changes from increasing to decreasing over northern South America and over southern and southeastern Asia. The corresponding change for 1941–1970 is similar in pattern to that for 2003–2022, but weaker in amplitude.

Lavers *et al.* (2023) have shown that 24-h ERA5 precipitation estimates accumulated from the final 4D-Var trajectories have generally smaller random errors than standard ERA5 estimates do when compared with station data for 2001–2020. Biases are larger in some regions, including Europe, associated with too much final-trajectory precipitation when little or none was observed. Figure 16b,d indeed shows decreasing precipitation over much of Europe for longer spin-up periods. The overall conclusion of the comparison with station data is nevertheless

TABLE 2 Contributions to global-mean evaporation and precipitation rates ($mm \cdot day^{-1}$) from oceanic regions averaged from 2003 to 2022, for six of the accumulations that can be constructed from archived European Centre of Medium-range Weather Forecasts Reanalysis v.5 data.

Accumulation	Effective spin-up	Evaporation	Precipitation
0–12 h final four-dimensional variational trajectory from 0900 and 2100 UTC	0–12 h	2.55	2.16
0–12 h forecast from 0000 and 1200 UTC analyses	3–15 h	2.54	2.25
0–12 h forecast from 0600 and 1800 UTC analyses (standard)	9–21 h	2.53	2.30
12–24 h forecast from 0000 and 1200 UTC analyses	15–27 h	2.51	2.29
24–36 h forecast from 0000 and 1200 UTC analyses	27–39 h	2.48	2.25
48–60 h forecast from 0000 and 1200 UTC analyses	51–63 h	2.46	2.26

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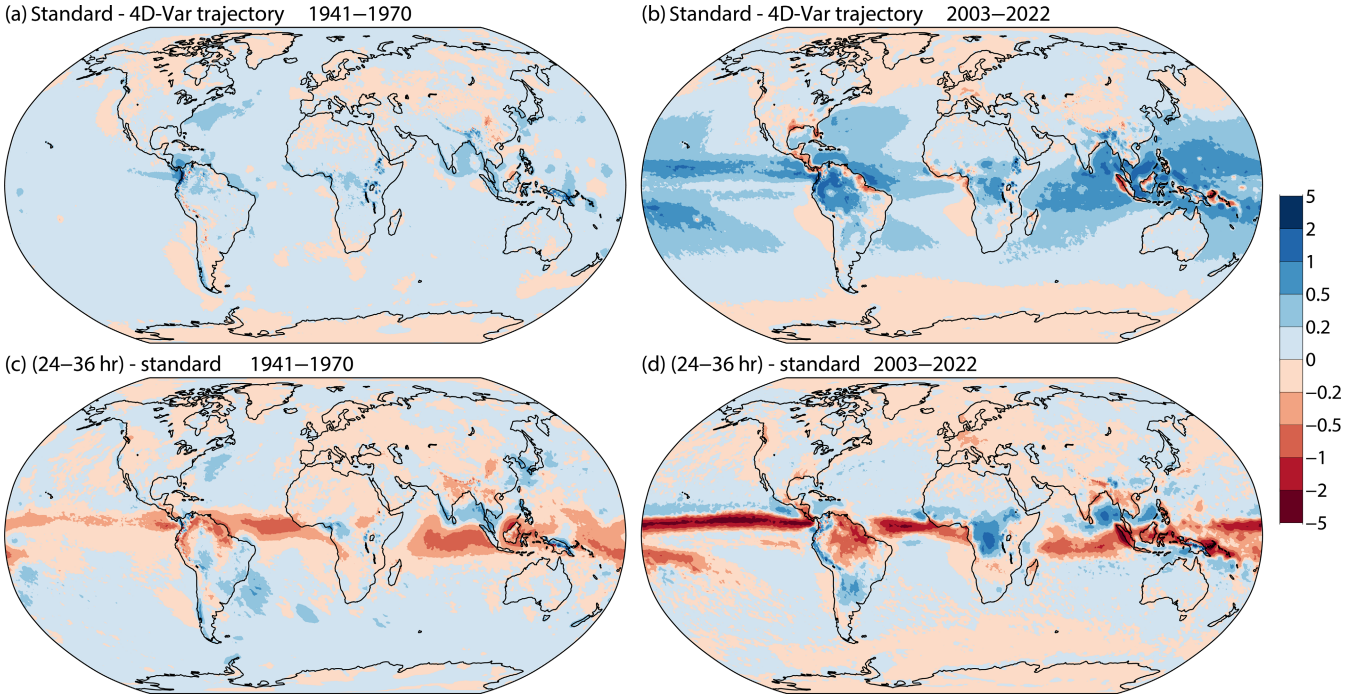

FIGURE 16 Differences in multiyear average precipitation rates (mm·day^{-1}) between (a) standard and four-dimensional variational (4D-Var) trajectory values for 1941–1970, (b) standard and 4D-Var trajectory values for 2003–2022, (c) 24–36 h forecast and standard values for 1941–1970, and (d) 24–36 h forecast and standard values for 2003–2022.

in favour of use of the final-trajectory values. The present results show that this would give a poorer global hydrological balance, but they also show that the apparently better balance between precipitation and evaporation of the standard short-range forecast estimates may be a rather fortuitous consequence of a global precipitation that first rises and then falls with increasing spin-up period. There remains a need for further assessment and understanding of this topic, which will be sought in the context of the revised modelling and data assimilation to be used for ERA6.

4.3 | Stratospheric water vapour

Stratospheric water vapour, like deep-layer soil moisture, is an ERA5 variable that is not influenced directly by observations. Although rapid and substantial variations can occur with disturbance and breakdown of the winter stratospheric polar vortex, such as illustrated for ERA5 by Hersbach *et al.* (2020), typical values of stratospheric water vapour may take several years to adjust following initialisation of a production stream. The better spun-up production streams of ERA5 thus bring some improvement over ERA5.P, although this improvement is substantial only for the first few years of the 1950s. Stratospheric water vapour also goes through an adjustment phase in the 1940s.

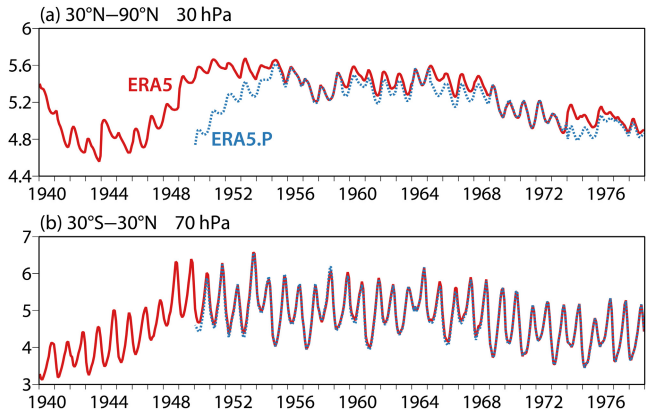

FIGURE 17 Time series from 1940 to 1978 of specific humidity (ppm) averaged monthly over (a) 30°N–90°N at 30 hPa and (b) 30°S–30°N at 70 hPa from ERA5 (red, solid) and ERA5.P (blue, dotted). ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; ERA5.P, ERA5 preliminary extension back to 1 January 1950.

Figure 17 provides an illustration. Figure 17a shows the variation over time from 1940 to 1978 of specific humidity averaged monthly and over the region north of 30°N, at the 30 hPa level. Water vapour is particularly slow to adjust to a changed initialisation in this region, and the behaviour for the level shown is similar to that at higher and lower levels, and similar to that south of 30°S, aside from larger

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annual cycles in the Southern Hemisphere, where condensation is triggered in the background forecasts by colder winter polar temperatures. Conversely, averages from 30°S to 30°N adjust relatively quickly to a changed initialisation in the lower stratosphere, such as at the 70 hPa level shown also in Figure 17b. Adjustment takes longer at higher levels in the tropical and subtropical stratosphere. This is a consequence of ERA5's representation of the slow transfer of water vapour by the Brewer–Dobson circulation, upward in the Tropics, poleward and downward in middle and high latitudes, and of the slow rate of methane oxidation in the lower and middle stratosphere.

The 1950s production stream for ERA5.P was initialised with lower values of stratospheric water vapour than are typical for the 1950s and 1960s, particularly in the extratropics. Here, adjustment to higher values takes some 5 years, as shown in Figure 17. The initial values from the other ERA5.P streams are much closer to the values from the better spun-up streams of ERA5, but small reductions in extratropical values that take several years to disappear occur when values from different streams are introduced in 1959, 1965, and 1974. Differences between ERA5.P and ERA5 are much smaller in the tropical and subtropical lower stratosphere, being barely visible in Figure 17b and of little concern given the challenges of representing well the transfer of water from the troposphere to the stratosphere and the subsequent upward movement of vapour.

The 1940s production stream for ERA5 was initialised with a representative value of extratropical stratospheric water vapour, but Figure 17 shows that water vapour declines here until the mid 1940s, after which it returns to more typical values. Water vapour in the tropical and subtropical lower stratosphere is initially atypically low, but it increases to more typical values by the late 1940s. This water vapour adjustment is almost certainly associated with adjustment of tropical tropopause temperatures over the same period. As at other atmospheric levels, these temperatures are biased low prior to the assimilation of sufficient radiosonde data, causing (in the assimilating model) excessive drying of air that enters the stratosphere in the Tropics and eventually reaches the extratropics.

5 | VARIABILITY OF TEMPERATURE FROM 1940 TO 2022

5.1 | Upper air temperature

Figure 18 presents time series related to global average temperatures at or near the 500 and 50 hPa levels of the atmosphere. It is similar to Bell *et al.* (2021, fig. 25), but in this case compares ERA5 with JRA-3Q rather than its predecessor JRA-55. The two figures are largely similar for the common period 1950–2020, in that there is little sensitivity of time series to the change from ERA5.P (not shown) to ERA5, and much of the discussion given by Bell *et al.* (2021) still applies. Figure 18b shows that ERA5 agrees better with JRA-3Q than with JRA-55 prior to about 1990 at 50 hPa, as JRA-3Q, like ERA5, cools a little faster than JRA-55 over this period. The converse is the case at 500 hPa, illustrated in Figure 18a, where (relative to 1981–2010) JRA-3Q is colder than ERA5 prior to 1980, more so than JRA-55 is.

The years 2021 and 2022 added here were both characterized by La Niña conditions over the tropical Pacific Ocean. Average global temperatures at 500 hPa were high in a multidecadal context but lower than those of several other years since 2015 and the earlier El Niño years of 1998 and 2010.

Figure 18e,f shows data fits for observations from the 450–550 hPa and 40–60 hPa layers rather than the 500 and 50 hPa levels shown by Bell *et al.* (2021). This is because the number of observations used by ERA5 from precisely 500 and 50 hPa has fallen in recent years. Instead, the assimilation system now chooses to use more data from nearby levels, which is possible because data from many radiosonde ascents are now being encoded and exchanged internationally with much higher vertical resolution (Ingleby *et al.*, 2016). Figure 18g,h illustrates how the overall number of observations assimilated for particular layers has increased since the mid 2010s. The rise is not a sudden one, as individual national meteorological services have implemented the necessary encoding change at different times, and not all have increased resolution initially. One consequence of this is that the background and analysis fits to observations shown in Figure 18e,f vary in recent years due to changes in the geographical distributions of data as well as due to any fundamental changes in background and analysis quality, as discussed by Hersbach *et al.* (2020).

Few midtropospheric radiosonde data are available to ERA5 until the beginning of 1946, when there is a jump in the number of assimilated temperature observations. A second jump in observation numbers occurs at the beginning of 1948, followed by a steady significant rise until 1958. The rise is much slower thereafter until the changes that begin in 2015. The ERA5 background model runs a little cold in the absence of assimilated observations (Hersbach *et al.*, 2015), but Figure 18a indicates that ERA5's global-mean 500 hPa temperature is at a reasonably consistent level from 1946 onwards as judged by comparison with JRA-3Q, notwithstanding systematic differences of a few tenths of a degree Celsius between the two global averages prior to 1980 (and larger regional differences, not shown, where data coverage is sparse). The mean differences between the observations and background ERA5 temperatures used are smaller than 0.2°C in magnitude

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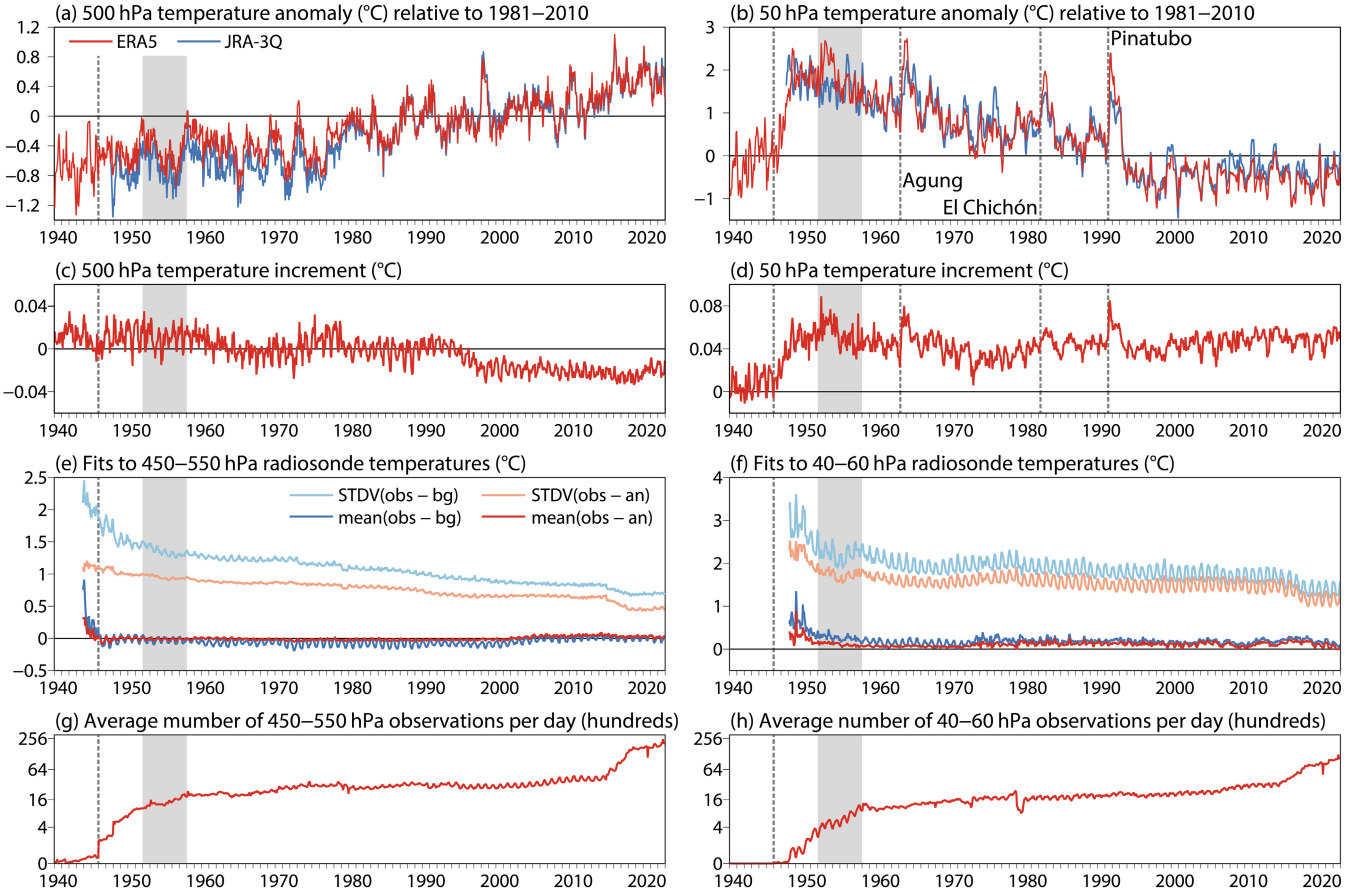

FIGURE 18 Time series from 1940 to 2022 of quantities related to global-mean temperature (°C) at (a, c, e, g) 500 hPa and (b, d, f, h) 50 hPa: (a, b) monthly mean anomalies with respect to 1981–2010 of ERA5 and JRA-3Q; (c, d) monthly mean ERA5 analysis increments; (e, f) monthly standard deviations and means of the ERA5 background and analysis departures for all radiosonde observations assimilated from the 450–550 hPa and 40–60 hPa layers; (g,h) monthly mean numbers of radiosonde temperature measurements per day for these layers. Grey shading denotes the years 1952–1957, and a vertical dashed line marks the start of 1946. The other vertical dashed lines in panels (b) and (d) mark the dates of the major volcanic eruptions of Mt Agung, El Chichón, and Mt Pinatubo. ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; GPCC, Global Precipitation Climatology Centre; JRA-3Q, Japanese Reanalysis for Three Quarters of a Century.

for all months of each year from 1946 onwards for the 450–550 hPa layer. The corresponding mean differences between observations and the ERA5 analyses are smaller than 0.1°C.

Temperature anomalies at 50 hPa (Figure 18b) during the very earliest period in ERA5 (1940–1945) are smaller than might be expected from the subsequent evolution of anomalies. This is due to a known model cold bias (Laloyaux *et al.*, 2020) affecting several ECMWF model cycles, including that used for ERA5 (CY41R2). During 1940–1945, the lack of upper air temperature observations exposes this model bias. During the latter part of the 1940s and early 1950s, as the number of upper air observations increases, the bias is corrected more effectively, as evidenced by the larger mean analysis increments (Figure 18d). This results in anomalies, by 1950, more consistent with those in subsequent years.

Only a small number of the 1946 and 1947 radiosonde ascents reach the lower stratosphere, but numbers for the vicinity of 50 hPa increase from 1948 to 1958, and global-average 50 hPa temperatures appear to be at a reasonable level from 1948 onwards as judged by comparison with JRA-3Q, apart from a difference from 1952 to 1954. The behaviour of ERA5 is in fact of special interest from 1952 to 1957, the period marked by grey shading in Figure 18. In ERA5, but not JRA-3Q, the lower stratosphere is warmer from 1952 to 1954 than for any earlier or later time, apart from a spell following the 1963 volcanic eruption of Mt Agung. There is a corresponding peak in the ERA5 analysis increment, suggesting that the higher temperatures are due to the assimilation of radiosonde temperatures that are relatively high. Peaks in the analysis increment also occur following the volcanic eruptions of Agung, Pinatubo in 1991, and, more weakly, El Chichón in

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1982. In these three cases substantial amounts of volcanic aerosol are prescribed in the assimilating model, following Coupled Model Intercomparison Project, Phase 5, and the increments are a sign that the model response is a warming that is weaker than indicated by the assimilated observations (Hersbach *et al.*, 2015). Much smaller amounts of volcanic aerosol, confined to the Northern Hemisphere, are specified for 1952–1954.

The difference between ERA5 and JRA-3Q at 50 hPa for this period is most likely due to differences in the assimilation of radiosonde data. This may include differences in the amounts of radiosonde data assimilated, differences in the quality control and bias corrections applied to these data, and differences in the relative weights given to the background and observed values in the data assimilation. The disagreement between the two reanalyses raises a question over the realism of ERA5's anomalous behaviour, but the suggestion from the aforementioned evidence that the ERA5 background model may have underestimated anomalous radiative warming of the lower stratosphere in the early 1950s is nevertheless supported by independent observational evidence for lower than usual amounts of solar radiation reaching the ground between 1952 and 1954, and for higher than usual amounts of stratospheric dust in 1953. Specific results and discussion, with references to other studies, are given by Enger and Fritz (1956) and Roosen and Angione (1984). Definitive attribution to a source or sources is lacking, however. Documented volcanic eruptions during the period appear to lack the required intensity, as indicated by the Coupled Model Intercomparison Project, Phase 5 forcing used in ERA5. Two major ground-burst nuclear tests were carried out over land during the period, on Enewetak Atoll in 1952 and Bikini Atoll in 1954.[1] Others in the 1954 series of tests were carried out over sea. The impact on atmospheric radiative heating of the material deposited in the tropical stratosphere by these explosions is not known, however.

Global-mean 500 hPa temperatures are relatively high in 1952 and 1953, but low from 1954 to early 1957. The 500 hPa increments shown in Figure 18c are a little lower in 1953 and 1954 than in adjacent years, but tropospheric variations appear to have been primarily forced in the ERA5 background forecasts by variations in SST. The absence of substantial variations in the increments indicates that the forced variations in 500 hPa temperature are largely consistent with the radiosonde observations ingested by ERA5.

5.2 | SST and marine air temperature

Figure 19a shows time series of SST and marine air temperature (MAT; temperature at a height of 2 m above the sea surface) from ERA5 averaged over all ice-free seas.

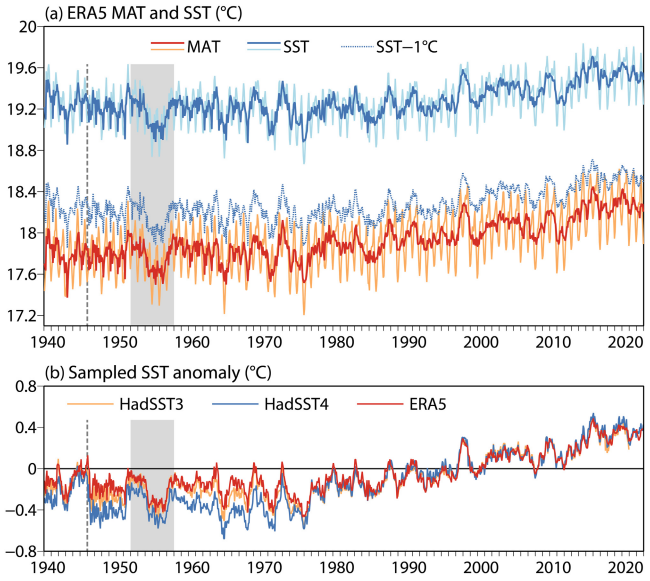

FIGURE 19 Time series of marine temperatures (°C) from 1940 to 2022: (a) monthly mean sea-surface temperature (SST) and marine air temperature (MAT; at 2 m) averaged over all ice-free seas, with and without removal of the mean annual cycle for 1981–2010, showing also the SST minus 1°C with annual cycle removed; (b) averages from 60°N to 60°S of SST anomalies relative to 1981–2010 for HadSST3, HadSST4, and ERA5 monthly means, using data only from the 5° × 5° grid boxes for which both HadSST3 and HadSST4 provide values. Grey shading denotes the years 1952–1957, and the vertical dashed line marks the start of 1946. ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; HadSST*x*, Hadley Centre Sea Surface Temperature version *x*.

The period from 1952 to 1957 has relatively high temperatures in 1952, 1953 and 1957, and a prolonged spell of below average temperatures from 1954 to 1956. A few individual months in the ERA5 data record are colder still, but the spell from 1954 to 1956 is the longest over which such atypically cold temperatures are sustained. This is the case for both SST and MAT. In fact, 1956 is the coldest calendar year in the ERA5 dataset for the globe as a whole.

Maps of annual-mean ERA5 surface air temperature anomalies relative to 1941–1970 are shown in Figure 20 for the years from 1952 to 1957. Temperatures over the tropical eastern Pacific Ocean show large swings, from patterns indicative of El Niño in 1953 to La Niña from 1954 to 1956, and back to El Niño in 1957. The La Niña was quite severe: the Oceanic Niño Index issued by the US National Weather Service's Climate Prediction Center shows larger minima only twice since then (https://origin.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ONI_v5.php). The two El Niños were less extreme.

Considerable interannual changes occur elsewhere. Cold conditions predominate over the oceans from 1954 to 1956, with mixed conditions away from the tropical

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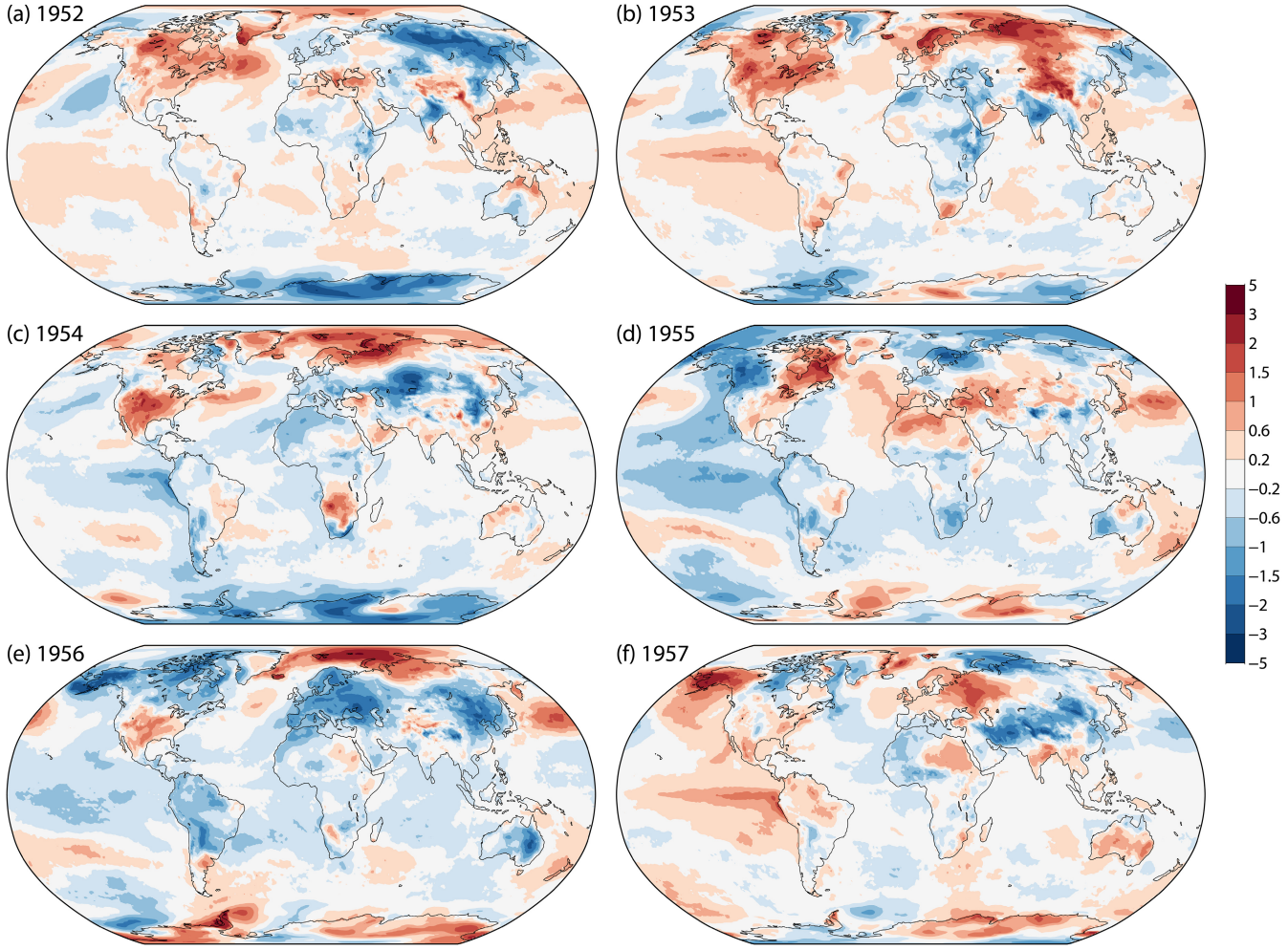

FIGURE 20 Anomalies of annual-mean European Centre of Medium-range Weather Forecasts Reanalysis v.5 2-m temperatures (°C) relative to 1941–1970 for (a) 1952, (b) 1953, (c) 1954, (d) 1955, (e) 1956, and (d) 1957.

Pacific in the other years. Variability over land is large over North America, Eurasia, and Antarctica. This is qualitatively similar to the longer term picture shown by Simmons (2022) for the variability of monthly means for 1979–2022 with linear trend removed. A similar picture is found for monthly means for 1940–1978 and, with weaker amplitude, for the variability of annual means for 1940–2022 (not shown).

In view of the degree of variability over both sea and land between 1952 and 1957 it is hard either to identify or to rule out a contribution to tropospheric cooling arising from the inferred changes in stratospheric composition discussed in Section 5.1.

Returning to Figure 19a, it can be seen that although on monthly and annual time-scales the variations of SST and MAT are highly correlated, the differences between the two decrease over time from 1940 to 2022. A tendency for decreasing differences has been noted previously by Simmons *et al.* (2017) for the ERA-Interim and JRA-55 reanalyses and earlier by Cowtan *et al.* (2015) for climate simulations. It has been discussed more recently by Gulev *et al.* (2021). It is a source of differences between the surface temperature trend estimates from reanalyses such as ERA5, which usually use MAT over sea, and the monthly climatological temperatures from datasets such as GISTEMP and Hadley Centre/Climatic Research Unit Temperature version 5 (HadCRUT5; Morice *et al.*, 2021) that use SST where sea is not frozen.

There is uncertainty over the size of the differences, however, as noted by Gulev *et al.* (2021). In the case of ERA5, ordinary least-squares linear trends of the monthly average SST minus MAT differences are for decreases of 0.022°C per decade over the period 1940–2022 and 0.019°C per decade for 1950–2022, but only 0.008°C per decade for 1979–2022. Though some decadal variability in differences is to be expected, it is likely that biases in the early decades of the reanalysis contribute to larger differences between SST and MAT then.

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There is also evidence that the SST analyses prescribed in ERA5 are biased high in the early decades, when they are taken from the Hadley Centre Sea Ice and Sea Surface Temperature dataset version 2 (HadISST2) dataset. Figure 19b shows time series of monthly average SSTs from the medians of the Hadley Centre Sea Surface Temperature version 3 (HadSST3; Kennedy *et al.*, 2011) and version 4 (HadSST4; Kennedy *et al.*, 2019) ensemble datasets and from ERA5. Data are expressed as anomalies relative to 1981–2010, which is the most recent reference period that can be used consistently for all time series of temperature anomalies presented in this section. HadSST3 and HadSST4 provide spatially incomplete estimates of SST anomalies at 5° × 5° spatial resolution. Averages for all three datasets are calculated using only values from grid boxes for which both HadSST3 and HadSST4 provide data, except for the final 2 years of the period, when ERA5 is sampled as HadSST4 and HadSST3 data are unavailable. The sampling appears to have little impact on the temporal variability of the average SST, as this is similar in Figure 19a,b. HadSST4 provides an exception, exhibiting a sharp drop in temperature in 1946—discussed by Kennedy *et al.* (2019)—that is larger than seen for the other datasets.

Only the sea-ice component of the HadISST2 dataset has been published (Titchner & Rayner, 2014), but it is clear from Figure 19b that the SST component is much closer to HadSST3 than it is to HadSST4. Furthermore, HadSST4 has generally lower temperatures than either of the other two datasets prior to the latter half of the 1970s. HadSST4 is now used as the source of marine temperatures in the Berkeley Earth (Rohde & Hausfather, 2020) and HadCRUT5 global datasets, which are two of the monthly climatological datasets often used together with ERA5 to depict global temperature trends. Two of the other such datasets, GISTEMP and NOAAGlobalTemp (Vose *et al.*, 2021) use Extended Reconstructed Sea Surface Temperature version 5 (ERSSTv5; Huang *et al.*, 2017) for marine temperature. Kennedy *et al.* (2019) provide justification for the changes from HadSST3 to HadSST4 and show how they bring closer agreement with ERSSTv5. ERA5 is accordingly a warm outlier over sea when compared with the four global temperature datasets that currently use either HadSST4 or ERSSTv5.

5.3 | Trend and low-frequency variability of surface air temperature over land

Figure 21 presents time series related to the analysis of surface air (2 m) temperature over all land surfaces and over the land area designated as Europe, which is a particular focus of monitoring by C3S. Relative to the 1981–2010 reference period, ERA5 exhibits a cold bias over land for the early years that tends to cancel its warm bias over sea when global averages are compared for these years. The cold bias is illustrated in Figure 21a, where temperature anomalies averaged over all land from ERA5 are compared with those from Berkeley Earth and GISTEMP, applying a 36-month running mean and selecting only two datasets other than ERA5 so that differences among these datasets are more easily seen. The bias is evident from 1940 to 1957, a period in which Berkeley Earth and GISTEMP are very close to each other and ERA5 is up to about 0.3°C cooler.

The three datasets are in good agreement for most of the period from 1958 onwards, but ERA5 again appears to be relatively cold during the mid to late 1970s. Corresponding C3S temperature monitoring plots presented at https://climate.copernicus.eu/climate-indicators/temperature show that, in this period, the JRA-55 reanalysis is close to ERA5, whereas two additional monthly climatological datasets, HadCRUT5 and NOAAGlobalTemp, are close to Berkeley Earth and GISTEMP. This difference in performance between the two types of analysis is largely accounted for by different averages over Antarctica.

Figure 21a also shows small but systematic differences after 2010. ERA5 warms slightly faster than Berkeley Earth and GISTEMP over land in recent years. In this case, however, differences are spread more generally among the datasets. GISTEMP is closer to ERA5 than it is to Berkeley Earth, and the C3S monitoring just referenced shows that HadCRUT5 and JRA-55 warm less than Berkeley Earth. NOAAGlobalTemp ranks between GISTEMP and Berkeley Earth.

Berkeley Earth and GISTEMP are the two datasets chosen for comparison with ERA5 in Figure 21 as they have higher horizontal resolution than HadCRUT5 and NOAAGlobalTemp. They were shown by Simmons *et al.* (2021) to be the closer to ERA5.P for averages over Europe, although rankings have changed following the release of a new version of NOAAGlobalTemp in early 2023. The averages over Europe presented in Figure 21b again show ERA5 to be relatively cold prior to 1967, when far fewer observations were available for use by its production streams (Figure 21h). Differences from ERA5 are larger for Berkeley Earth than GISTEMP, and are larger still for HadCRUT5, as shown by C3S monitoring. The latest version of NOAAGlobalTemp again ranks between GISTEMP and Berkeley Earth. In all cases, however, the differences among datasets are small compared with the variations in European temperatures that occurred during the period. In particular, ERA5 captures the extremity of the cold temperatures that prevailed over Europe in the early 1940s, for which further discussion is given in subsequent subsections.

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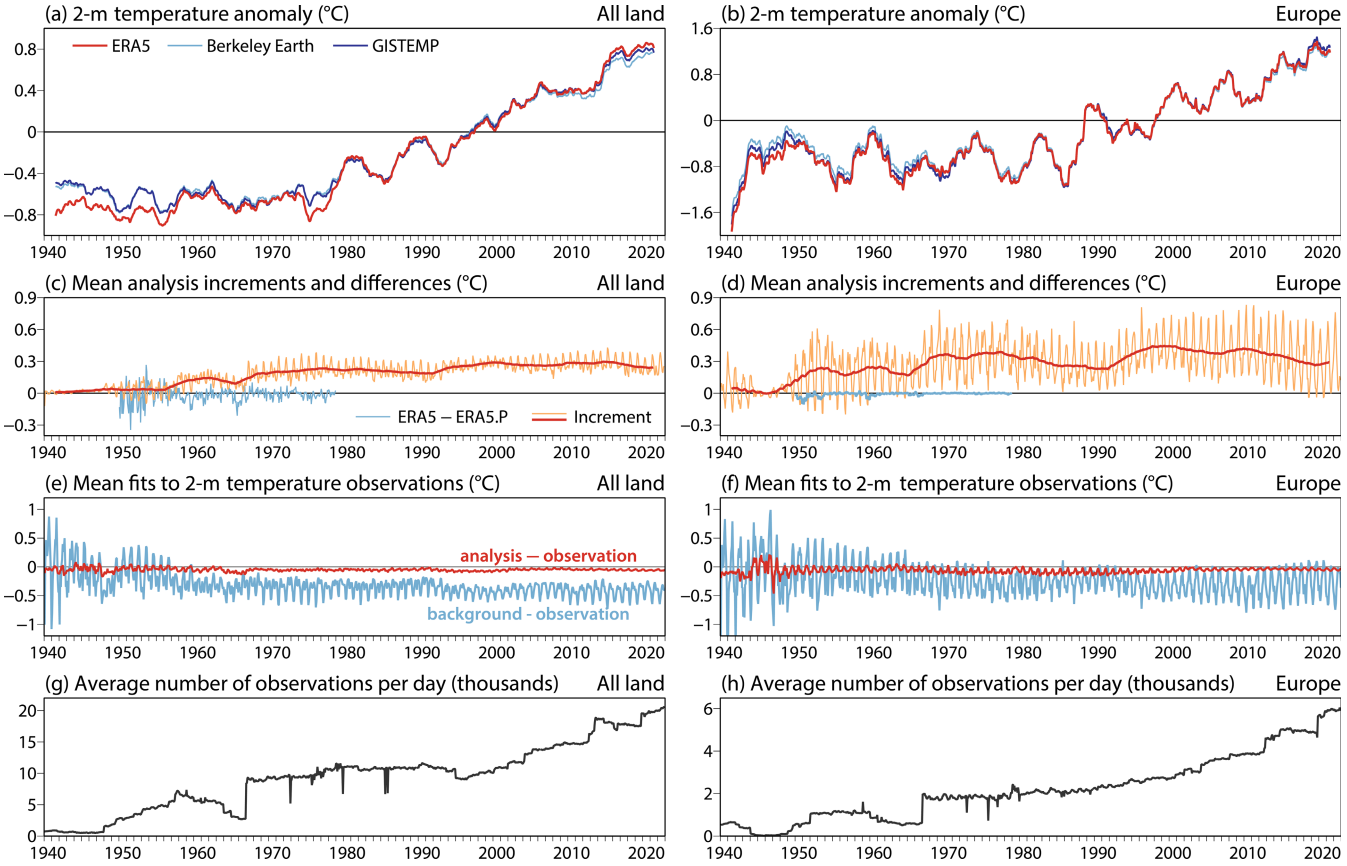

FIGURE 21 Time series from 1940 to 2022 related to 2-m temperature over (a, c, e, g) all land and (b, d, f, h) Europe (as defined in the caption of Figure 11) showing (a, b) 36-month running means of monthly temperature anomalies relative to 1981–2010 from ERA5, Berkeley Earth, and GISTEMP; (c, d) monthly and 36-month running mean ERA5 analysis increments (analysis minus background differences), and monthly mean differences between ERA5 and ERA5.P; (e, f) monthly mean (background minus observation) and (analysis minus observation) differences averaged over all observations for which feedback is available from the surface analysis; and (g, h) monthly means of the number of observations per day for which feedback is available. ERA5, European Centre of Medium-range Weather Forecasts Reanalysis v.5; ERA5.P, ERA5 preliminary extension back to 1 January 1950; GISTEMP, Goddard Institute for Space Studies Surface Temperature Analysis.

The other panels of Figure 21 relate more directly to the performance of the ERA5 analysis system. They show time series of analysis increments, the differences between ERA5 and ERA5.P, the fits of background and analysis to observations, and the numbers of temperature observations analysed for which feedback on data usage has been archived. The latter amount to about one-third of the observations available in the early decades of ERA5, but a smaller fraction later on, when observations become more frequent due to automation (Simmons *et al.*, 2021). The change from ERA5.P to ERA5 lowers temperatures in the early 1950s, for European as well as global averages, and also has a large cooling impact in 1965 and 1966, as illustrated earlier for Africa, in Figure 12. Prior to 1967, when observation numbers are relatively low, the magnitude of the increments evidently depends on the number of observations. Where observations are present the analysis is close to them, faltering mainly over Europe from 1944 to 1949 when coverage is at its poorest. Background minus observation differences are fairly steady from the 1960s onwards.

There is, nevertheless, a small shift in the background minus observation differences in the early 1990s. This is accompanied by a shift in the increments, which keeps the analyses close to the observations. The shift in the background signifies that a change in one or more of the observation types assimilated by ERA5 has changed the performance of the background model. One candidate is the assimilation from the early 1990s of significant numbers of rain-sensitive microwave radiances over sea, which may influence background temperatures over land through changes in the global hydrological cycle.

Background minus observation differences are generally negative. Following Simmons *et al.* (2004), this has usually been interpreted as evidence of a cold bias of the background, a bias that is corrected by the analysis of observations. Though this seems to have been largely justified in the past, improvements in the forecasting

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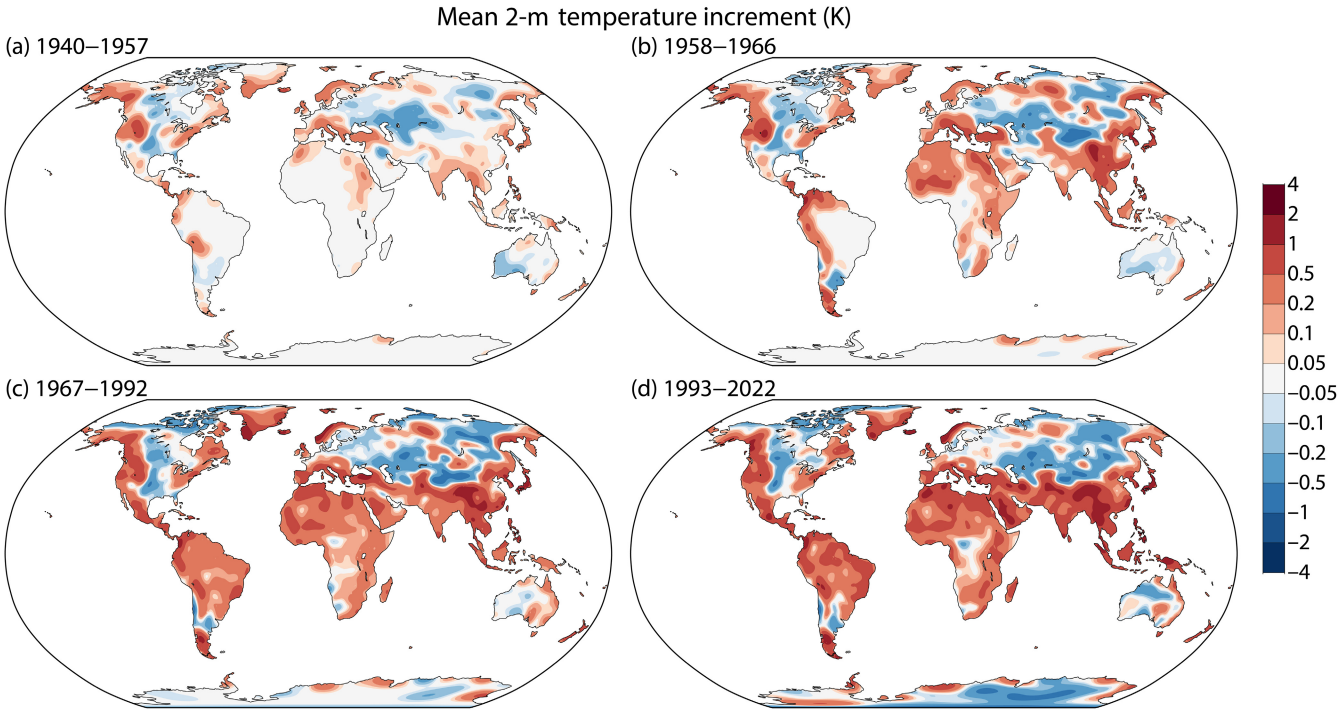

FIGURE 22 Analysis increments (analysis minus background differences) for ERA5 2-m temperatures (°C) averaged for (a) 1940–1957, (b) 1958–1966, (c) 1967–1992, and (d) 1993–2022.

system have revealed that a significant portion of the remaining bias is, in fact, attributable to a warm bias of the analysis. This is because of a failure to account for a bias in the representativity of the observations, especially in mountainous regions, due to a tendency for observing stations to be located at altitudes that are lower than average terrain heights. The feedback data for ERA5 include station altitudes and the heights of the ERA5 orography at the station locations. Analysis of these data reveals an average height mismatch of approximately 50 m when averaged globally for the observations used in recent years. This equates to a temperature bias of 0.3°C for a lapse rate of 6°C·km^{-1}, given that the surface fields from ERA5 are intended to be representative of values at the height of ERA5's model terrain. It should be noted that the 50 m average mismatch is probably an underestimate of the full mismatch, as it is based only on observations that are used in the analysis. Observations that have mismatches of more than 300 m are not used.

Whether such an error is present in the background rather than the analysis, or the analysis rather than the background, does not matter for time series that cover only the period since 1967, when observational coverage is reasonably good. Figure 21 dates back to 1940, however, and displays values as anomalies relative to 1981–2010. What appears to be a cold bias of ERA5 in the 1940s and 1950s may thus be at least partly due to a warm bias of the 1981–2020 climatological reference.

Figure 22 presents global maps of the ERA5 analysis increments for 2-m temperature averaged over different segments of the reanalysis period. In each of the four periods shown there are warming increments over most of the main mountain chains: over the Rockies and Andes, and over Norway, southern Europe, the Middle East and southern Asia. These regions, plus parts of central and northeastern Asia, are those that contribute most to the average difference between station and model altitudes. Amplitudes are weaker for 1940–1957, and to a lesser extent 1958–1966, than they are after 1967. This is not only due to the smaller number of observations used: it is found that the average difference between station and model altitudes is some 10–20 m smaller for observations from the 1940s than it is for observations from recent decades.

Away from the mountainous regions just noted, there are cooling increments over central North America, central and northern Asia, and, in recent years, the Antarctic plateau. Warming increments dominate in the Tropics and Subtropics, and it is here that the small increase from 1967–1992 to 1993–2022 is most evident.

5.4 | Monthly anomalies in surface temperature

Maps of surface temperature anomalies for two months from the early 1940s are shown in Figure 23 for four

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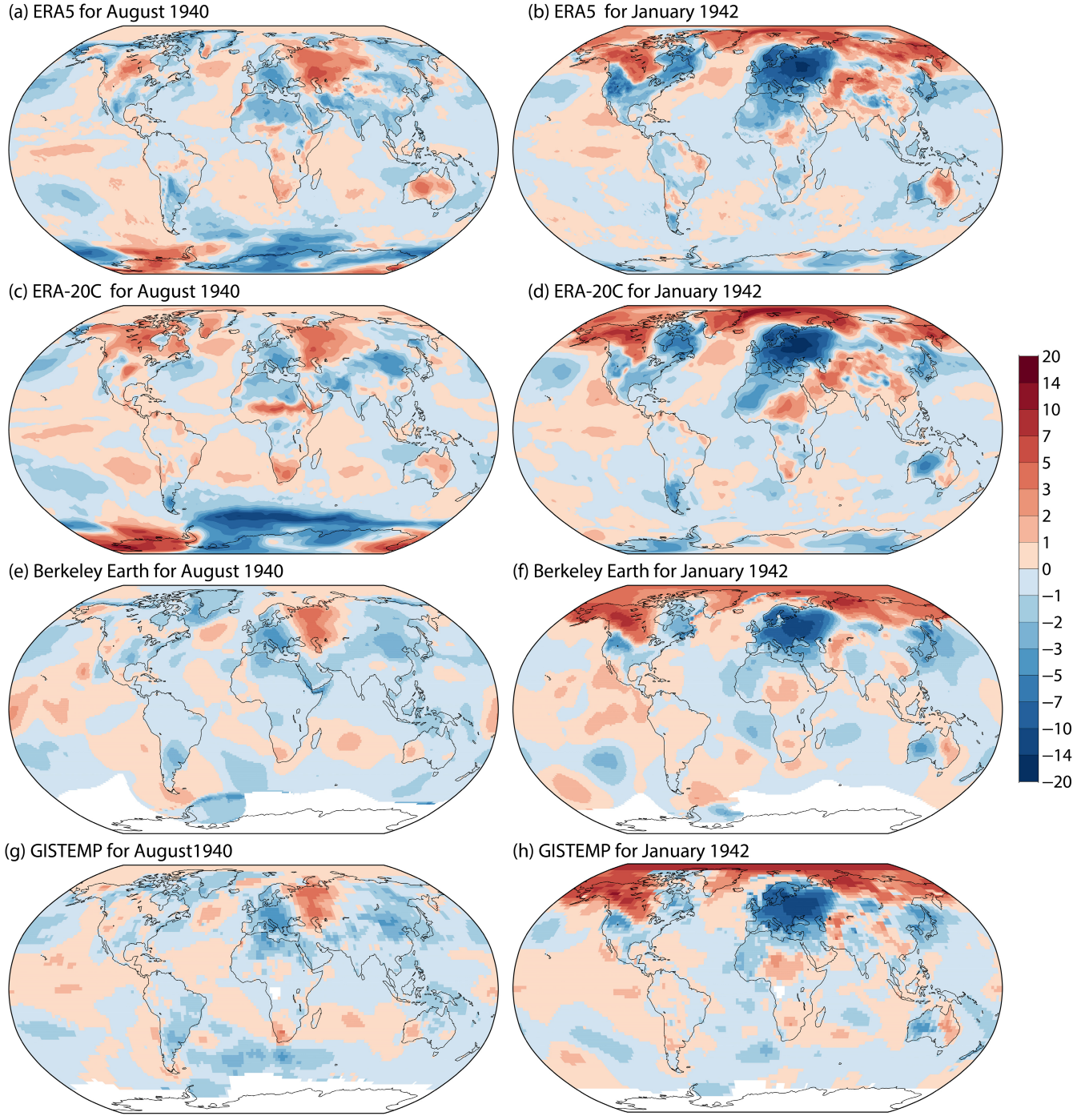

FIGURE 23 Anomalies of monthly-mean two-metre temperatures (°C) relative to 1981–2010 for (a, c, e, g) August 1940 and (b, d, f, h) January 1942 from (a, b) ERA5, (c, d) ERA-20C, (e, f) Berkeley Earth and (g, h) GISTEMP. ERA5, European Centre of Medium-range Weather Forecasts (ECMWF) Reanalysis v.5; ERA-20C, ECMWF Reanalysis of the 20th Century; GISTEMP, Goddard Institute for Space Studies Surface Temperature Analysis.

datasets. The months are August 1940, when it was unusually warm over western Russia, and January 1942, the coldest month of all for Europe in the ERA5 data record, although in this regard the month is difficult to separate from February 1956—illustrated by Bell *et al.* (2021) for ERA5.P—because of changes in bias over time. Anomalies are shown for ERA5, ECMWF Reanalysis of the 20th Century (ERA-20C; Poli *et al.*, 2016), Berkeley Earth, and GISTEMP. ERA-20C was chosen for inclusion to demonstrate that much of the capability of ERA5 for these early years comes from its assimilation of surface pressure and marine surface wind data. These were the only data assimilated directly by ERA-20C. ERA5 assimilated these data and not much more for the early

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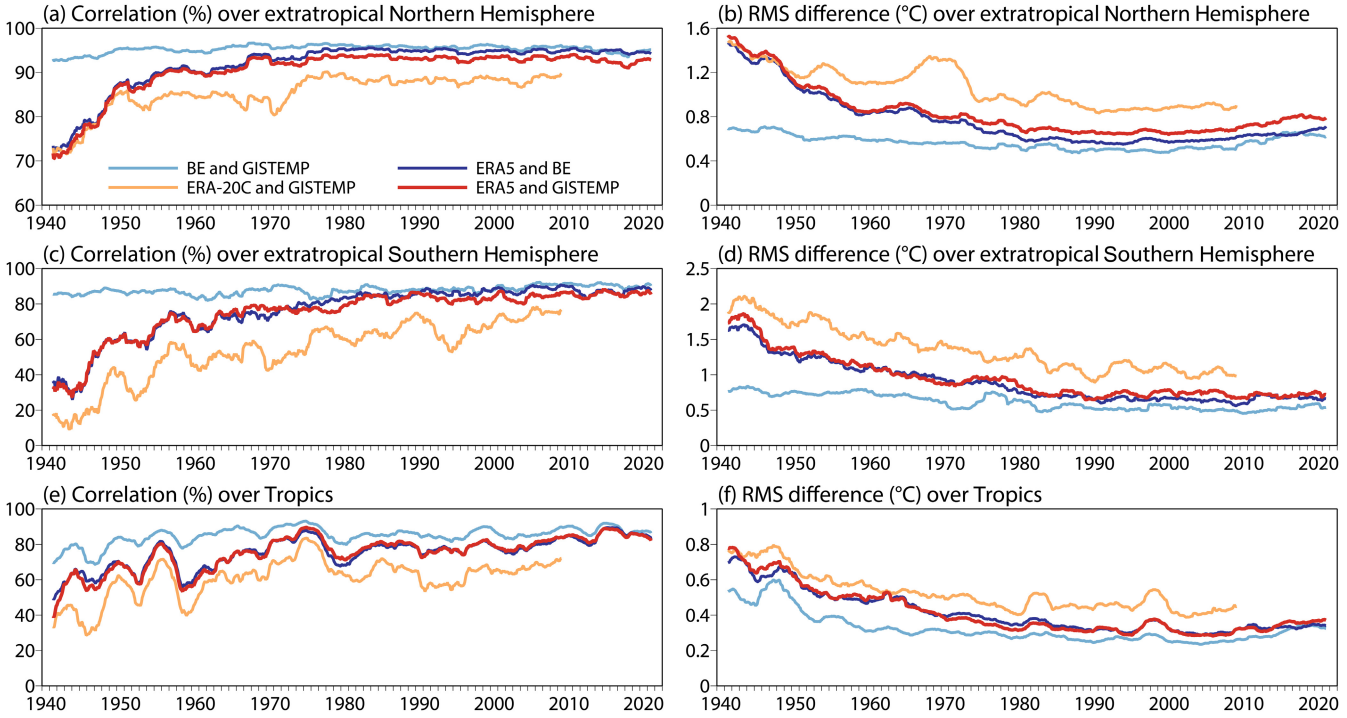

FIGURE 24 (a, c, e) Correlations (%) and (b, d, f) root-mean-square (RMS) differences (°C) between pairs of ERA5, ERA-20C, Berkeley Earth (BE), and GISTEMP 2-m temperature anomalies relative to 1981–2010 for running 36-month periods from 1940 to 2022, over land areas of the (a, b) extratropical Northern Hemisphere, (c, d) extratropical Southern Hemisphere, and (e, f) Tropics. Calculations are carried out at 2° × 2° resolution using in all cases data only from grid boxes for which both Berkeley Earth and GISTEMP provide complete data coverage. This excludes Antarctica and large parts of the Congo Basin and Borneo from the calculations. ERA5, European Centre of Medium-range Weather Forecasts (ECMWF) Reanalysis v.5; ERA-20C, ECMWF Reanalysis of the 20th Century; GISTEMP, Goddard Institute for Space Studies Surface Temperature Analysis.

1940s. ERA-20C runs only to the year 2010, and the anomalies in Figure 23, like others in this section, are relative to 1981–2010. The choice of reference period is not critical as the main monthly anomalies are much larger in magnitude than decadal shifts in climatological averages.

It is evident from Figure 23 that the major temperature anomalies over Europe and elsewhere around middle latitudes of the Northern Hemisphere are similar for the four datasets. The main exception in this regard is ERA-20C, which is alone in the extent of its warm conditions over Canada for August 1940. Also pronounced for ERA-20C this month is the warm band south of the Sahara Desert, something present to a lesser extent in ERA5 but not in the other two datasets. Behaviour here is similar to what is shown in Figure 12 for the versions of ERA5 over the data-sparse 1965–1966 period. More generally, agreement between the two reanalyses and between the reanalyses and the other datasets is quite limited in the Tropics and extratropical Southern Hemisphere early in the reanalysis period.

Increasing agreement over time is illustrated in Figure 24, which presents time series of correlations and root-mean-square differences between the anomalies from ERA5 and Berkeley Earth, ERA5 and GISTEMP, ERA-20C and GISTEMP, and Berkeley Earth and GISTEMP. Values are shown for the extratropical Northern and Southern Hemispheres and for the Tropics for overlapping 36-month periods to filter out short-term variability.

Berkeley Earth and GISTEMP employ broadly similar methods of analysis, and for some periods they utilize monthly climatological temperature observations for regions where ERA5 does not have access to historical synoptic temperature data (Bell *et al.*, 2021). Figure 24 shows that from the 1940s to the 1970s GISTEMP and Berkeley Earth are distinctly closer to each other than either is to the ERA analyses. There is, nevertheless, some reduction over time in the differences between Berkeley Earth and GISTEMP during this period. There is little change in the levels of agreement after then, and little difference between the levels of agreement between GISTEMP and either Berkeley Earth or ERA5. It is not straightforward to attach significance to the small differences that remain in the latter period, as the Berkeley Earth data are smoother spatially than the GISTEMP data used here.

As seen for the examples in Figure 23, the datasets agree much better in the extratropical Northern Hemisphere than in the Tropics and Southern Hemisphere

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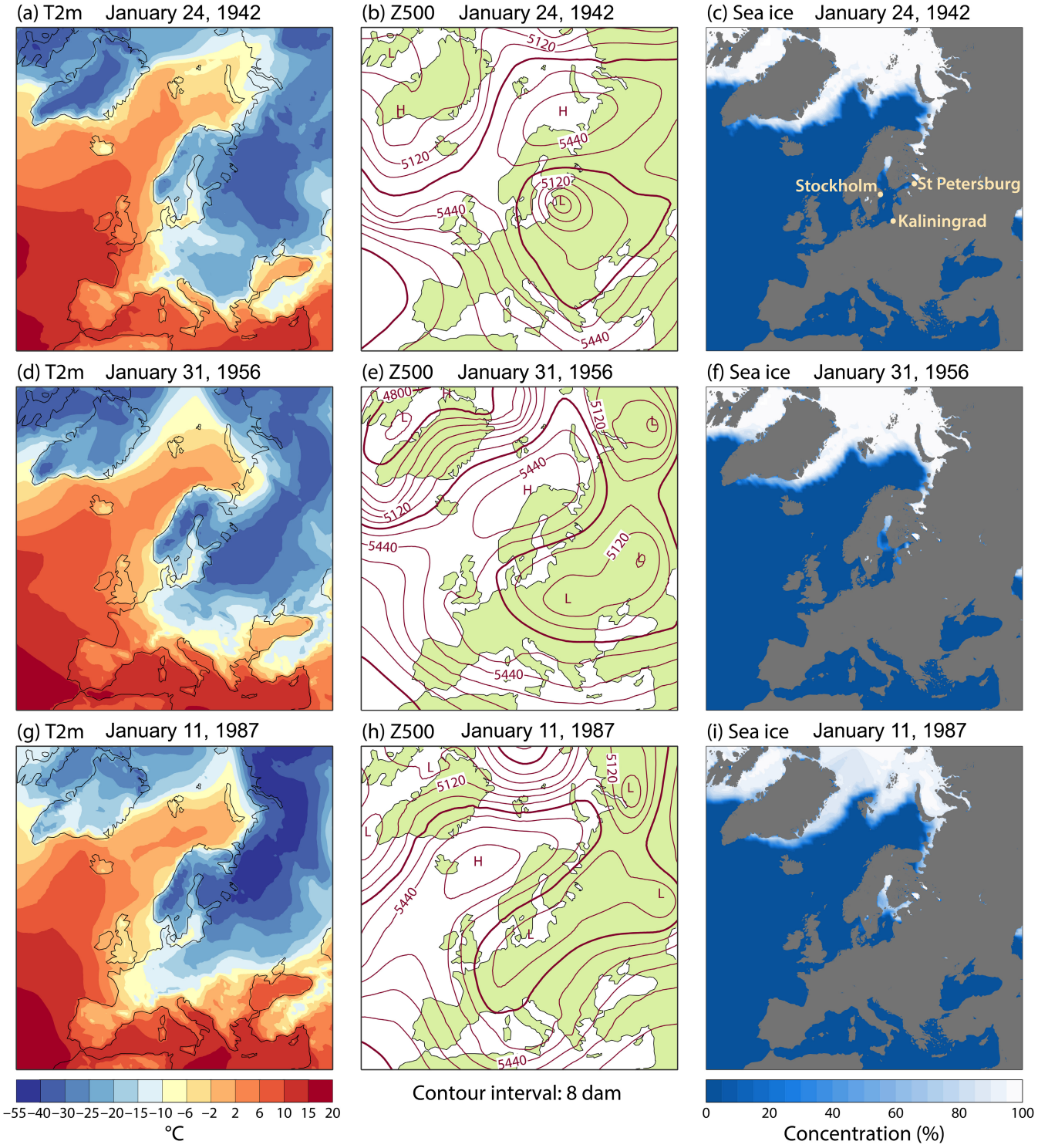

FIGURE 25 (a, d, g) Daily average 2-m temperature, (b, e, h) 1200 UTC 500 hPa geopotential height, and (c, f, i) sea-ice concentration (right) from European Centre of Medium-range Weather Forecasts Reanalysis v.5 for (a–c) 24 January 1942, (d–f) 31 January 1956, and (g–i) 11 January 1987.

early in the period. There is little to distinguish ERA5 from ERA-20C initially in the Northern Hemisphere, but ERA5 is the closer of the two to GISTEMP and Berkeley Earth from the late 1940s onwards. ERA5 is the closer virtually from the outset in the Tropics and Southern Hemisphere, where data coverage is initially very sparse and ERA5 presumably benefits from its better modelling and data assimilation.

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5.5 | Cold winter extremes in surface air temperature

Daily average 2-m temperatures over Europe from 1 January 1940, to 31 December 2022, have been examined to find the most extreme days for the continent as a whole. The three years with the coldest days are 1942, 1956, and 1987. 20–25 January 1942, 31 January and 9–11 February 1956, and 8–11 January 1987, are 14 of the 15 coldest days in the ERA5 record. Figure 25 presents maps focused on Europe showing the daily average temperatures for the coldest days of each of the three years, 24 January 1942, 31 January 1956, and 11 January 1987. The 500 hPa geopotential heights and sea-ice concentrations are also shown.

Lejenäs (1989) presented a synoptic analysis of observations for the spell in January 1942. He noted that the January 1987 spell was similar. In both cases, and also at the end of January 1956 as depicted by ERA5, the coldest days were associated with a predominantly easterly component to the flow that spread cold air from Russia, with blocking high pressure over the Norwegian and Barents seas, and low pressure to the south. A pronounced cyclonic vortex was present in 1942.

The 2-m temperatures at three locations (labelled in Figure 25c) have been examined. ERA5 had access to temperature observations from the immediate vicinity of only one of the three, Kaliningrad, for January 1942. Lejenäs (1989), quoting an earlier study by Scherhag, reported that the −34°C observed at Kaliningrad in the morning of 25 January 1942, was the lowest value in the area since January 1813. ERA5 had a minimum temperature of −35°C at the model grid-point (54.66°N, 20.62°E) nearest to the station currently designated to be Kaliningrad (54.70°N, 20.62°E), and analysed observations as low as −35°C from a location specified to be (54.78°N, 20.62°E) and −34°C from nearby (54.95°N, 20.25°E). Lejenäs (1989) reported a minimum of −31°C for St Petersburg 1 day earlier, a value seen also for the nearest ERA5 grid point. He also reported a minimum of −28°C for Stockholm on 25 January but in this case the nearest ERA5 grid point had temperatures that day that did not fall below −18°C.

One factor that likely contributed to the too-high ERA5 temperatures for Stockholm during the 1942 cold spell is the absence in ERA5 of the high concentrations of ice over the Baltic Sea reported by Lejenäs (1989). The HadISST2 sea-ice analyses used by ERA5 prior to 1979 do not represent the variability that can be seen in the yearly maximum Baltic ice extents displayed online by the European Environment Agency at https://www.eea.europa.eu/ims/arctic-and-baltic-sea-ice. These data, based on analyses by the Finnish Meteorological Institute, show that the largest ice extents in the ERA5 period occur in the years 1940, 1942, and 1947, followed by 1987 and 1956. HadISST2 has no year in the 1940s with high sea-ice concentrations, and although its typical values shift from 1952 to 1953 coincident with a change in data source, interannual variability remains low prior to a second change in data source from 1971 to 1972.

In contrast, ERA5 used sea-ice analyses for January 1987 (Eastwood *et al.*, 2014) that utilized scanning multichannel microwave radiometer data and had much higher concentrations of ice in the Baltic, particularly in the gulfs of Bothnia and Finland. Background air temperatures benefited more generally from the assimilation of considerably more data, and the corresponding 2-m temperature analyses had the further benefit of access to many more direct observations of surface air temperature. The temperature of the ERA5 grid point nearest Stockholm is below −20°C throughout the four days 9–12 January 1987, in agreement with what was seen then at the Stockholm Observatory for the first time since observations began there in 1756 (Lejenäs, 1989). Averages of the 0900 and 2100 UTC temperature observations made for the four days at the city's Bromma and (more distant) Arlanda airports are −23.0 and −24.0°C respectively. They were fitted by ERA5 with average (observation minus analysis) differences of 0.5 and 0.0°C respectively. Average background differences are −0.8 and −1.3°C. All temperatures available to ERA5 from Arlanda and Bromma are below −20°C throughout the four days apart from the 1200 UTC observations from Bromma on 11 and 12 January which are −19.8 and −19.4°C respectively. The nearest-grid-point temperatures from ERA5 for St Petersburg and Kaliningrad reach as low as −37 and −25°C respectively during this period. Corresponding observed minima are −34 and −25°C.

ERA5 produced nearest-grid-point temperatures below −30°C for both St Petersburg and Kaliningrad for 31 January 1956, in accord with the observations it used. Its lowest grid-point value for Stockholm, −19°C, is about 2°C higher than observed. Temperatures interpolated to station locations are a further 2°C higher. Lack of Baltic sea-ice may again be a factor.

5.6 | Hot summer extremes in surface air temperature

In contrast with the timing of the cold winter extremes, Europe's hottest summer days, as measured by the average ERA5 temperature over the continent, have been in recent years. Four spells, in July–August 2010, July–August 2018, July 2019, and August 2020, provide the 21 hottest days in the 1940–2022 ERA5 record, and the next hottest day was in August 2022. Indicative as this is of a warming European

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climate, it is nevertheless worthy of note that the hottest three European days in the part of the 20th century covered by ERA5 were as long ago as July 1941 and August 1947. Moreover, the anomalously high monthly temperatures over European Russia illustrated in Figure 23 for August 1940 included a day with then-record high temperatures over the south of the country that were surpassed only in July 2010.

The routine monitoring of temperatures by C3S that has been in place since August 2015 provides several examples of summer extremes in its monthly, seasonal, and annual reports, and occasional feature articles. They are available at https://climate.copernicus.eu and are not repeated here. They cover both European and global extremes. An article on the warmth of Siberia in June 2020 (https://climate.copernicus.eu/arctic-siberias-unusual-warm-spell-continues) notes that ERA5 recorded its highest ever June temperature in the Arctic, 37°C, on the 20th of that month. On the same day, a temperature of 38°C was reported by the nearby observing station at Verkhoyansk, a value now confirmed by the World Meteorological Organization to be a new Arctic temperature record (https://public.wmo.int/en/media/press-release/wmo-recognizes-new-arctic-temperature-record-of-38%E2%81%B0c), based in part on the consistency between the observation and ERA5.

The success or otherwise of ERA5 in capturing such temperature extremes depends in part on whether the extreme is dependent on local topographical conditions that the assimilating model is unable to resolve well. This may explain why the highest ERA5 temperature in the vicinity of Lytton, in western Canada, was a little over 43°C on 29 June 2021, some 6°C lower than the reported observation from Lytton. In contrast, the highest temperature analysed at ERA5 model grid points close to Coningsby in the UK was 40.4°C on 19 July 2022, very close to the new national temperature record of 40.3°C set there on that day (https://www.metoffice.gov.uk/about-us/press-office/news/weather-and-climate/2022/record-high-temperatures-verified).

6 | SUMMARY AND CONCLUSIONS

The latest version of ECMWF's global atmospheric reanalysis, ERA5, has been extended to 1940 and continues to run in near real time. The resulting reanalysis, currently spanning 83 years, is the longest "full-input" reanalysis produced to date. The accuracy of ERA5, as indicated by reforecast skill, exceeds that of its predecessor ERA-Interim. Indeed, in the Northern Hemisphere, the step change in skill from ERA-Interim to ERA5 generally exceeds that of any of the previous generations of ECMWF reanalyses. The significant reduction in quality further back in time, especially prior to the mid 1950s in the Northern Hemisphere and prior to the late 1970s in the Southern Hemisphere, continues to motivate ongoing activities to recover hitherto unexploited observations during the first half of the 20th century and satellite data from the early satellite era starting in the mid 1960s.

The production of this revised extension to ERA5 offered the opportunity to address several problems with the preliminary version that spanned 1950–1978, as well as to extend the reanalysis to 1940, around the beginning of the upper air observational record. Improvements were made to the bias correction of surface pressure observations and the assimilation of IBTrACS observations. The latter, involving adjustments to the quality control and assigned observation errors, resulted in more realistic analyses of tropical cyclones and wave heights. The more optimal initialisation of the new production streams, either from the preliminary version or a dedicated low-resolution scout run, resulted in more continuous evolution of soil moisture and stratospheric humidity.

The improved initialisation of soil moisture in particular resulted in more realistic representations of several episodes of anomalously high surface temperatures and low precipitation estimates over Africa and South America in the 1950s and 1960s. The long spin-up times for stratospheric humidity resulted in several discontinuities, or seams, between production streams in the preliminary extension that were significantly improved in the final version.

Several interesting aspects are evident in the temperature anomaly time series in the lower stratosphere. First, ERA5 provides an informative case study of the impact of model biases on reanalyses, in this case a tendency of the model to run cold in the lower stratosphere, an effect exacerbated by the absence of observational constraints. The reanalysis mean state exhibits erroneously cold temperatures in the layer centred on 50 hPa through the early 1940s, a bias that is only corrected as the number of radiosonde observations increases through the late 1940s and early 1950s. Second, a warm anomaly that qualitatively resembles similar anomalies associated with the later volcanic eruptions of Agung, Pinatubo, and El Chichón is evident from 1952 to 1954. Possibly associated with this is a notable cooling of marine surface air temperatures from 1954 to 1956. It is speculated that this may be a result of land-based nuclear tests during 1952–1954, which, most likely, injected significant amounts of aerosol and dust into the lower stratosphere.

Trends in 2-m temperature anomalies over land for ERA5, both globally and over Europe, show generally excellent agreement with independent datasets based entirely on surface observations, particularly for the late

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1970s onwards. There are episodes earlier in the ERA5 record when differences are larger, at several tenths of a degree Celsius. This aspect of the work has highlighted a component of the bias in ERA5 analyses related to the mismatch between the altitude of observing stations and their altitude as represented in the model, which results in a warm bias of up to 0.3°C when averaged over the global set of observations. It is expected that this feature of the assimilation can be improved in future reanalyses. Root-mean-square differences between ERA5 and these observation-based estimates (of anomalies) are lower than 1°C in the Northern Hemisphere/Southern Hemisphere from the mid 1950s/mid 1960s and correlations exceed 90%/70% for the same periods. An analysis of the fidelity of ERA5 in representing the coldest extremes (mainly in the first half of the period covered) and warmest extremes (generally in the last two decades) over Europe throughout the whole record demonstrates the growing credibility of reanalyses in verifying temperature extremes, but it has also highlighted the importance of considering the impact of local topographic detail.

As lessons learned for future reanalyses, several general points are noteworthy: the need for effective monitoring in both model space and observation space to identify problems at the earliest opportunity; the need for further development work on the treatment of biases in both observations and the forecast model; the optimal initialisation of variables that are slow to adjust in the reanalysis, and; the ever-continuing need for the recovery and reprocessing of observational data. Current efforts are being directed at improving all of these aspects in advance of the next-generation ECMWF reanalysis, ERA6, due to start production in 2024.

ACKNOWLEDGEMENTS

Reanalysis touches on a large number of activities at ECMWF, and its success relies on the efforts from many, many people across ECMWF and from many collaborations. This study has been partially funded by the Copernicus Climate Change Service. ECMWF implements this Service and the Copernicus Atmosphere Monitoring Service on behalf of the European Commission. We wish to thank the two reviewers whose suggestions and constructive comments led to improvement of the manuscript.

CONFLICT OF INTEREST STATEMENT

To our knowledge there is no conflict of interest of the material presented in this article.

DATA AVAILABILITY STATEMENT

The ERA5 data that support the findings of this study are openly available from the C3S Climate Data Store (e.g., https://doi.org/10.24381/cds.f17050d7; https://doi.org/10.24381/cds.6860a573).

ENDNOTE

[1]Dates were respectively 1 November and 1 March local time, but 31 October and 28 February UTC.

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